A Theory of Firm Boundaries and Knowledge Sharing*

Evgenii Fadeev† Michael Powell‡

June 3, 2026

The latest version is here

Abstract

We study asset ownership as an instrument of knowledge protection in collaborations where production requires sharing private knowledge but makes parties vulnerable to non-contractible expropriation. Ownership confers control over production and the ability to expropriate. It can protect the owner's knowledge by reorganizing production so that certain parties need not be informed, or by ensuring that some parties who must be informed lack the means to expropriate — allowing information to be shared safely. However, because ownership also gives the owner the ability to expropriate, other parties may withhold their knowledge from the owner. We apply the theory to explain the role of third-party intermediaries that facilitate knowledge sharing among competitors, and to determine when a vertically integrated firm must spin off a subsidiary to encourage knowledge sharing from rivals.

*We are grateful for comments from Charles Angelucci, Ashish Arora, Daniel Barron, Wouter Dessein, Bob Gibbons, Oliver Hart, Andrew Koh, Niko Matouschek, Andrea Prat, Andrei Shleifer, Alexander Wolitzky, and participants at various conferences and seminars.

†Duke University, Fuqua School of Business, email: evgenii.fadeev@duke.edu.

‡Northwestern University, Kellogg School of Management, email: mike-powell@kellogg.northwestern.edu.

1 Introduction

Firm boundaries are often drawn to protect proprietary knowledge (Teece, 1980; Liebeskind, 1996). Consistent with this view, recent empirical work highlights acquisitions as a mechanism for facilitating knowledge transfers (Demirer and Karaduman, 2025) and suggests that such transfers may be a primary purpose of ownership (Atalay, Hortaçsu, and Syverson, 2014). This paper studies when and why firm boundaries shape the transfer and use of knowledge.

As a motivating example, consider General Motors' (GM) decision to spin off its parts division, Delphi Automotive, in 1999. We argue that integration protected GM's knowledge but discouraged rival automakers from working with Delphi "for fear that it would share parts designs and other competitive information with G.M." (Bradsher, 1998). Spinning off Delphi was a way to make it a more attractive supplier to GM's rivals, but only by giving up some of the knowledge protection that integration gave GM.

Our argument builds on three premises. First, the use of knowledge is *non-contractible*: once knowledge is shared, it cannot be taken back, and it is difficult to restrict how the recipient uses it. The recipient may expropriate it for private gain at the expense of the original holder, for example by applying it to competing products or developing competing follow-on technologies. This feature of knowledge distinguishes it from tangible inputs (Arrow, 1962).

Second, knowledge must be combined with a productive asset such as a specialized plant or equipment. Although parties cannot contract on how knowledge is used, they can allocate ownership of the asset, and hence the *control rights* that govern its use. These control rights can protect the owner's knowledge in two ways: (i) they allow the owner to apply its knowledge directly without disclosing it, and (ii) when disclosure is necessary, they allow the owner to prevent a partner from using the asset to engage in knowledge expropriation.

Finally, ownership confers *residual* rights of control associated with the asset — that is, *all* rights not specified in a contract (Grossman and Hart, 1986). This set of rights is a bundle: some of these rights protect the owner, while others expose non-owners to expropriation. In the GM–Delphi example, GM's ownership allowed it to prevent expropriation by Delphi, but it also gave GM control over production decisions related to other automakers that relied on Delphi. As a result, serving a rival required GM to receive the rival's proprietary knowledge and, if informed, GM could expropriate it.

We develop a model in which parties collaborate on projects. Each project specifies a collection of tasks. A task consists of an action and some target party's privately observed state and is completed if the action matches that state. Projects exhibit O-ring complementarities (Kremer, 1993): a project succeeds and generates value only if all tasks are completed. Performing these tasks may require parties to receive information about others' states. But if they are sufficiently informed, they may attempt to expropriate that information. A single productive asset is required for the completion of certain tasks, and ownership confers control over the actions involved in those tasks. Some parties may also need the asset to engage in expropriation, and ownership allows them to do so.

We begin by analyzing this setting through the lens of information design (Bergemann and Morris, 2019). Given an ownership structure, an omniscient mediator chooses a decision rule that maps states into private action recommendations. The mediator's goal is to maximize total surplus. The information conveyed by the recommendations determines how precisely parties learn about others' states. This approach establishes a benchmark that isolates how the non-contractible use of knowledge limits total surplus.

Our main analysis focuses on the case in which expropriation is sufficiently costly that it is optimal to deter it: a party capable of expropriation must not receive enough information about another party's state to make expropriation profitable. For each task, this requirement places an upper bound on how precisely information about the target's state can be disclosed to the acting party. We refer to this upper bound as the task's *safety threshold*.

Our first main result characterizes an optimal decision rule for any fixed ownership structure. Whenever this decision rule recommends a correct action for a low-threshold ("hard") task, it also recommends correct actions for all higher-threshold ("easier") tasks. This nested property is optimal because tasks are complementary within a project, so additional precision on an easier task is valuable only in states where the harder tasks are also completed. As a result, each project is constrained by its *bottleneck*: the task with the lowest safety threshold.

This characterization implies that ownership matters only through its effect on project bottlenecks. Changing ownership changes a project's bottleneck by altering who must be informed for this project to succeed and who can expropriate if informed. Our second main result shows that surplus-maximizing ownership structures maximize the value-weighted sum of bottleneck safety thresholds across projects.

We use this result to derive several implications for ownership patterns. First, if two parties are sufficiently tempted to expropriate each other's knowledge, third-party ownership may be optimal. Common suppliers and other intermediaries can act as *knowledge hubs*: rather than share information directly, competitors often outsource the decisions that combine their information to a third party that is less tempted to expropriate it (Küpper et al., 2020).

Second, when the value of one project is sufficiently large relative to others, the optimal ownership structure relaxes the bottleneck in that project. As in our motivating example, when GM's business initially dominated, it was optimal to keep Delphi as a subsidiary to maximize the value of GM's business. As rivals' business grew in value (Helper and Henderson, 2014), the cost of withholding their knowledge rose until the spin-off became optimal.

Extensions. We conclude by considering several variations of the model. First, we show that the optimal decision rule and ownership structure can be supported as an equilibrium outcome in a game where parties contract over ownership, transfers, and communication through a neutral uninformed mediator. In this setting, the mediator conditions recommendations on parties' reports rather than on the true state, so parties can strategically misreport their information. This equivalence clarifies that the central friction in our environment is the non-contractibility of actions rather than private information.

Second, we show that it is the non-contractibility of expropriation that gives rise to our main trade-offs. When expropriation actions are fully contractible, parties can achieve the first best (the full value of all projects with no expropriation) regardless of ownership. More generally, partial contractibility of expropriation — through payments conditioned on an imperfect signal of expropriation — increases total surplus. In contrast, allowing for full contractibility of actions in project tasks or the division of project values does not improve total surplus.

Third, without a mediator, parties can still deter expropriation through cheap talk but cannot coordinate information disclosure across tasks. When coordinated disclosure is unnecessary (as in the spin-off application), the optimal ownership structure is unchanged. When coordinated disclosure matters (as in the knowledge-hub application), the inability to coordinate may shift optimal ownership, suggesting a role for neutral intermediaries — such as consulting or law firms — in raising total surplus.

Fourth, bundling control rights into a single asset implies that the owner must be informed about another party's state for a project to succeed — information the owner can then expropriate — so information is optimally withheld. We show that if control rights can be sufficiently unbundled across multiple assets, the first best is achievable.

Finally, our main analysis assumes prohibitive expropriation costs, so our optimal decision rule deters expropriation entirely. When costs are not prohibitive, it can be optimal to tolerate some expropriation to enable greater disclosure.

Related Literature. We build on the incomplete-contracts approach to firm boundaries (Grossman and Hart, 1986; Hart and Moore, 1990; Hart, 1995). Grossman and Hart (1986) define ownership as the allocation of residual rights of control: when specific rights are too costly to enumerate contractually, ownership assigns the rights left unspecified. Much of this literature focuses on the allocation of control. We follow it in this respect but also emphasize the *residual* nature of ownership: it typically transfers *multiple* decision rights as a bundle.

Within the firm-boundaries literature, our paper is closest methodologically to work that studies ownership structures through a mechanism-design approach (Klemperer, Cramton, and Gibbons, 1987; Matouschek, 2004; Segal and Whinston, 2016; Baliga and Sjöström, 2018). Whereas these papers emphasize how ownership affects information rents when ex post actions are contractible, we study optimal ownership when ex post actions are non-contractible. The central issue is therefore about obedience rather than truth-telling.

To characterize the optimal information disclosure and ownership structure, we use modern information-design tools (Rayo and Segal, 2010; Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Mathevet, Perego, and Taneva, 2020). This approach to study organizational-design questions has also been taken by Kolotilin and Li (2021), who study relational communication in repeated relationships with voluntary transfers and show that selective pooling can be used to manage obedience constraints. All these papers take control structures as given.

A related literature studies how information structures and mediated recommendations shape cooperation in cartels. Sugaya and Wolitzky (2018) show that coarser observability can make collusion easier to sustain because finer information lets firms tailor profitable deviations to market conditions. Marshall and Marx (2007) and Ortner, Sugaya, and Wolitzky (2024) study mediated arrangements that help parties collude, in part through private recommendations and sometimes through correlated randomization. In our paper, related instruments serve as building blocks for a theory of the firm.

We also connect to the literature on communication and delegation in organizations (Dessein, 2002; Alonso, Dessein, and Matouschek, 2008; Rantakari, 2008), which studies how the allocation of control shapes communication. Because ours is a paper on firm boundaries rather than internal organization, we emphasize two features of the environment that delegation models typically abstract from: parties have access to rich contracting instruments (i.e., they can contract jointly on decision rights, information-sharing protocols, and transfers), and the relevant choice is ownership, which allocates a bundle of inextricable control rights.

Finally, our paper relates to the literature on selling information. A central theme in this work is that an informed party faces appropriation hazards when sharing knowledge, and various mechanisms can mitigate them: leveraging competition and credible threats of wider disclosure (Anton and Yao, 1994, 2002), or gradual revelation that relaxes enforcement or informational constraints (Hörner and Skrzypacz, 2016; Garicano and Rayo, 2017; Fudenberg and Rayo, 2019). In our setting, ownership provides an alternative mechanism: it can eliminate appropriation hazards for the owner, but at the cost of creating new ones for others.

2 Model

There are N parties, $i \in \mathcal{N} = \{1, \ldots, N\}$, who collaborate on a collection of projects. Each project succeeds only if a set of required actions is correctly tailored to private information held by the parties. Sharing information improves cooperation but creates a risk: parties may use what they learn to expropriate others' knowledge for their private benefit.

Information. The state is $\theta = (\theta_1, \dots, \theta_N)$, where $\theta_i \in \{-1, 1\}$ is privately observed by party i. The components of θ are independent, and each realization is equally likely, $p(\theta) = \left(\frac{1}{2}\right)^N$.

Actions and Asset Ownership. Each party $i \in \mathcal{N}$ chooses a vector of actions. We distinguish actions by whether they are (i) *cooperative* or *expropriation*, and (ii) *alienable* or *inalienable*. Cooperative actions are productive: when correctly matched to the relevant component of the state, they contribute to project success. Expropriation actions are privately beneficial but socially destructive: matching them to others' information generates a private gain for the acting party while imposing a cost on others. Alienable actions require access to a productive asset, whereas inalienable actions reflect specific human capital and require no such asset.

There is a single productive asset owned by party $g \in \mathcal{N}$. We model control over alienable actions in the following way. Each party $i \in \mathcal{N}$ selects a plan for *alienable cooperative actions*, denoted $x_{ij} \in \{-1, 1\}$ for all $j \in \mathcal{N}$, and for *alienable expropriation actions*, denoted $z_{ij} \in \{-1, 0, 1\}$ for all $j \neq i$. While every party specifies a plan, only the choices of the asset owner g can be payoff-relevant.

Each party $i \in \mathcal{N}$ also chooses *inalienable cooperation actions*, denoted by $\bar{x}_{ij} \in \{-1, 1\}$ for all $j \in \mathcal{N}$, and *inalienable expropriation actions*, denoted by $\bar{z}_{ij} \in \{-1, 0, 1\}$ for $j \neq i$.

For each party $i \in \mathcal{N}$, let $a_i = ((x_{ij})_{j \in \mathcal{N}}, (\bar{x}_{ij})_{j \in \mathcal{N}}, (z_{ij})_{j \neq i}, (\bar{z}_{ij})_{j \neq i})$ denote party i's *action plan*. Let $\mathcal{A}_i$ denote the set of possible action plans available to party i and write $\mathcal{A} = \prod_{i \in \mathcal{N}} \mathcal{A}_i$ and $a = (a_i)_{i \in \mathcal{N}} \in \mathcal{A}$. Equivalently, we may write $a = (x, \bar{x}, z, \bar{z})$.

Some parties can expropriate only if they own the asset, while others can expropriate even if they do not. Let $\mathcal{E} \subseteq \mathcal{N}$ denote the set of parties whose expropriation opportunities are tied to the asset. We define party i's effective expropriation action against party j as ζ_{ij}. For parties who can intrinsically expropriate ($i \notin \mathcal{E}$), $\zeta_{ij} = \bar{z}_{ij}$. For parties requiring the asset ($i \in \mathcal{E}$), $\zeta_{ij} = z_{ij}$ for the asset owner ($g = i$), and $\zeta_{ij} = 0$ for non-owners ($g \neq i$).

Projects. Production is organized into a set of projects $\mathcal{K}$. Each project $k \in \mathcal{K}$ has value $V_k \geqslant 0$ if successfully implemented and value 0 otherwise. Project success requires that a set of cooperative actions be correctly matched to the relevant components of the state.

Formally, each project k has two sets of requirements. First, $\mathcal{R}_k \subseteq \mathcal{N}$ is the set of alienable requirements: for each $j \in \mathcal{R}_k$, the asset owner g must set $x_{gj} = \theta_j$. Second, $\bar{\mathcal{R}}_k \subseteq \mathcal{N} \times \mathcal{N}$ is the set of inalienable requirements: for each $(i, j) \in \bar{\mathcal{R}}_k$, party i must choose $\bar{x}_{ij} = \theta_j$. Given asset owner g, define the project-success indicator $w_k \in \{0, 1\}$ by

$$w_k(x, \bar{x}, g; \theta) = \left(\prod_{j \in \mathcal{R}_k} \mathbf{1}\{x_{gj} = \theta_j\} \right) \cdot \left(\prod_{(i,j) \in \bar{\mathcal{R}}_k} \mathbf{1}\{\bar{x}_{ij} = \theta_j\} \right).$$

Project k succeeds if and only if all of its alienable and inalienable requirements are met.

Payoffs. Party i's payoff is given by $u_i(a, g; \theta) + t_i$, where t_i is a monetary transfer and

$$u_i(a, g; \theta) = u_i^c(x, \bar{x}, g; \theta) + u_i^e(z, \bar{z}, g; \theta).$$

The term u_i^c represents the *cooperative payoff* from project success, and u_i^e represents the *expropriation payoff*. Transfers satisfy the budget-balance condition, $\sum_{i \in \mathcal{N}} t_i = 0$. We next describe the cooperative and expropriation payoff terms.

The cooperative payoff depends only on which projects succeed. We assume party i receives an exogenous share $\alpha_{ik} \geqslant 0$ of project k's value, with $\sum_{i \in \mathcal{N}} \alpha_{ik} = 1$. Its cooperative payoff is

$$u_i^c(x, \bar{x}, g; \theta) = \sum_{k \in \mathcal{K}} \alpha_{ik} V_k w_k(x, \bar{x}, g; \theta).$$

Thus, cooperative payoffs depend on actions only through project success and do not depend directly on asset ownership beyond its effect on which actions are payoff-relevant.

Expropriation payoffs depend on the effective expropriation actions ζ_{ij}. If party i attempts to expropriate party j's knowledge, and the attempt is successful (i.e., $\zeta_{ij} = \theta_j$), then party i obtains a private benefit $b_{ij} > 0$, while party j incurs a loss $\lambda_{ij} > 0$. If the attempt is unsuccessful (i.e., $\zeta_{ij} = -\theta_j$), party i incurs a cost $c_{ij} > 0$. If $\zeta_{ij} = 0$, both parties receive 0. Party i's expropriation payoff is therefore

$$u_i^e(z, \bar{z}, g; \theta) = \sum_{j \neq i} [b_{ij} \mathbf{1}\{\zeta_{ij} = \theta_j\} - c_{ij} \mathbf{1}\{\zeta_{ij} = -\theta_j\}] - \sum_{j \neq i} \lambda_{ji} \mathbf{1}\{\zeta_{ji} = \theta_i\},$$

where $\zeta_{ij} = \zeta_{ij}(z_{ij}, \bar{z}_{ij}; g)$ is the effective expropriation action.

We define total surplus as $TS(a, g; \theta) := \sum_{i \in \mathcal{N}} u_i(a, g; \theta)$. Because transfers are budget-balanced, they do not directly enter this expression.

We impose the following two assumptions on expropriation payoffs:

Assumption 1. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $\lambda_{ij} - b_{ij} > V_{\text{total}} := \sum_{k \in \mathcal{K}} V_k$.*

Assumption 2. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $b_{ij} < c_{ij}$.*

The first assumption ensures that parties jointly prefer to avoid expropriation. The second assumption ensures expropriation is risky and thus attractive only when a party has sufficiently precise information about the target's state.

Non-Contractibility of Actions. Following Arrow (1962), we assume that once knowledge is shared, its subsequent use is difficult to restrict contractually. We capture this friction by assuming that actions are *non-contractible*.

Decision Rule. Following the information-design-in-games approach of Bergemann and Morris (2019), we consider an *omniscient* mediator who decides how information is disclosed to the parties. A *decision rule* is a mapping $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$ for each $\theta \in \Theta$, which assigns to each realized state θ a distribution over action profiles $a = (a_i)_{i \in \mathcal{N}}$.

Timing. Fix an ownership structure $g \in \mathcal{N}$. The timing is as follows: (i) the mediator commits to a decision rule σ; (ii) the state θ is realized; (iii) each party i observes θ_i; (iv) the mediator draws an action profile a according to $\sigma(\cdot \mid \theta)$ and privately recommends a_i to each party i; (v) the parties choose their actions; and (vi) payoffs are realized.

Obedience. Recommendations from the mediator are not enforceable. We restrict attention to *obedient* decision rules, under which no party wishes to deviate from the recommended action after observing its private recommendation and private state. We denote the set of obedient decision rules by Σ. For any fixed ownership structure, the obedience constraints characterize the set of Bayes correlated equilibrium outcomes, where an outcome is a joint distribution over actions and states, $\psi(a, \theta) := p(\theta)\sigma(a \mid \theta)$.

Solution Concept. For a fixed ownership structure $g \in \mathcal{N}$, we find an *optimal decision rule* that maximizes total surplus subject to obedience. Let

$$TS(g) := \max_{\sigma \in \Sigma} \mathbb{E}\left[TS(a, g; \theta)\right], \tag{2.1}$$

where the expectation is taken with respect to the joint distribution $\psi(a, \theta) := p(\theta)\sigma(a \mid \theta)$. An *optimal ownership structure* is any g^* that solves $\max_{g \in \mathcal{N}} TS(g)$.

Discussion of Assumptions. In Section 5.1, we show that the outcome identified by our solution concept can be supported as an equilibrium outcome in a broader contracting game.

The model also relies on several assumptions that we relax in Section 5. We discuss how the results change under the contractibility of actions and project shares (Section 5.2), unmediated cheap-talk communication (Section 5.3), separable control rights across multiple assets (Section 5.4), and arbitrary expropriation costs (Section 5.5).

3 Analysis

We solve the model in two steps. First, we characterize the optimal decision rule for a fixed ownership structure. We show that it is optimal to ensure that no party with the means to expropriate receives enough information to profitably engage in expropriation. Second, we characterize the optimal ownership structure. We show that ownership determines both who must be informed and who has the means to expropriate.

3.1 Optimal Information Disclosure

Fix an ownership structure $g \in \mathcal{N}$ and a decision rule $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$. The mediator draws an action profile $a = (a_1, \ldots, a_N)$ from $\sigma(\cdot \mid \theta)$ and privately recommends a_i to each party i. Party i observes its private state θ_i and the recommendations in a_i. The decision rule is obedient ($\sigma \in \Sigma$) if all parties prefer to follow their recommended actions:

$$\sum_{\theta_{-i} \in \Theta_{-i}} \sum_{a_{-i} \in \mathcal{A}_{-i}} p(\theta_{-i} \mid \theta_i) \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big((a_i, a_{-i}), g; \theta_i, \theta_{-i}\big) - u_i\big((a_i', a_{-i}), g; \theta_i, \theta_{-i}\big) \Big] \geqslant 0$$
$$\text{for all } i \in \mathcal{N},\ \theta_i \in \Theta_i,\ a_i \in \mathcal{A}_i,\ a_i' \in \mathcal{A}_i. \qquad (3.1)$$

The set of inequalities in (3.1) defines obedience constraints in cooperative and expropriation actions. As we show below, the main results are driven by expropriation obedience constraints, which can be characterized using posterior beliefs about states. For $j \neq i$, let

$$\tau_{ij}(\theta_i, a_i) := \mathbb{P}_\psi\big(\theta_j = 1 \,|\, \theta_i, a_i\big)$$

denote the posterior belief of party i that $\theta_j = 1$ after observing (θ_i, a_i). This posterior is defined under the joint distribution $\psi(\theta, a) = p(\theta)\sigma(a|\theta)$. When the dependence on (θ_i, a_i) is clear from context, we write τ_{ij} for $\tau_{ij}(\theta_i, a_i)$.

Suppose party i can engage in expropriation. Then it chooses $\zeta_{ij} = 1$ if $\tau_{ij} b_{ij} - (1 - \tau_{ij}) c_{ij} > 0$, chooses $\zeta_{ij} = -1$ if $(1 - \tau_{ij}) b_{ij} - \tau_{ij} c_{ij} > 0$, and chooses $\zeta_{ij} = 0$ otherwise. Therefore, if party i receives a recommendation not to expropriate ($\zeta_{ij} = 0$), its posterior belief must satisfy

$$\max\{\tau_{ij}, 1 - \tau_{ij}\} \leqslant \tau_{ij}^\star := \frac{c_{ij}}{b_{ij} + c_{ij}},$$

where we assume that parties do not expropriate when indifferent. Assumption 2 ($c_{ij} > b_{ij}$) implies that $\tau_{ij}^\star > 1/2$.

Before stating the general results on the structure of an optimal decision rule, it is useful to build intuition through two examples. The first example shows that it is optimal to disclose information to a party capable of expropriation only up to the point at which expropriation is deterred. The second example shows that it is optimal to correlate recommendations.

Example 1: Optimal Decision Rule For One Party. Suppose there are two parties, $\mathcal{N} = \{1, 2\}$, and one project, $\mathcal{K} = \{1\}$. The project does not require an asset, $\mathcal{R}_1 = \varnothing$, but requires party 1 to match the state of party 2 via its inalienable action, $\bar{\mathcal{R}}_1 = \{(1, 2)\}$. That is, the project succeeds if and only if $\bar{x}_{12} = \theta_2$. Parties do not require the asset to engage in expropriation, $\mathcal{E} = \varnothing$. Suppose both parties receive half of the project's value, $\alpha_{11} = \alpha_{21} = 1/2$.

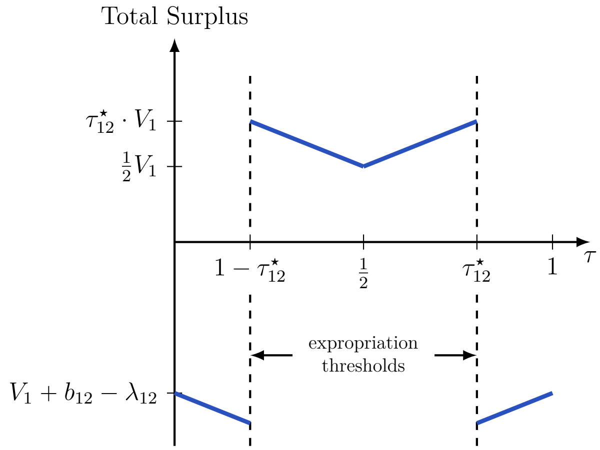

Figure 1: Total Surplus As a Function of Beliefs (Example 1)

This figure illustrates the relationship between total expected surplus and party 1's belief that $\theta_2 = 1$ for Example 1.

In this example, the asset ownership is irrelevant.

The optimal decision rule can be found using Bayesian Persuasion argument (Kamenica and Gentzkow, 2011). Denote by τ the posterior belief of party 1 that $\theta_2 = 1$ after receiving the recommendation from the mediator. Figure 1 illustrates the expected total surplus as a function τ. It is optimal for party 1 to select $\bar{x}_{12} = 1$ if $\tau > 1/2$ and $\bar{x}_{12} = -1$ if $\tau < 1/2$. If party 1's belief about θ_2 is sufficiently precise, it also optimally engages in expropriation. Specifically, if $\tau > \tau^\star_{12}$, party 1 selects $\zeta_{12} = 1$, and $\tau < 1 - \tau^\star_{12}$, party 1 selects $\zeta_{12} = -1$.

Under expropriation, the total expected surplus is negative. Thus, the optimal decision rule is to randomize between posteriors $\tau^\star_{12}$ and $1 - \tau^\star_{12}$. One way to construct such a belief distribution, which will be useful in the general construction, is as follows. Suppose there is a random variable $\xi \sim \text{Unif}[0, 1]$ independent of θ. The recommendations to party 1 are

$$x_{12} = \bar{x}_{12} = \begin{cases} \theta_2 & \text{if } \xi \leqslant \tau^\star_{12}, \\ -\theta_2 & \text{if } \xi > \tau^\star_{12}, \end{cases} \quad \text{and} \quad z_{12} = \bar{z}_{12} = 0.$$

Example 2: Nested Decision Rule For Multiple Parties. Suppose there are three parties, $\mathcal{N} = \{1, 2, 3\}$, and two projects, $\mathcal{K} = \{1, 2\}$. Neither project has alienable requirements, so $\mathcal{R}_1 = \mathcal{R}_2 = \varnothing$. Project 1 requires two inalienable actions from parties 1 and 2 to match the state of party 3, so $\bar{\mathcal{R}}_1 = \{(1, 3), (2, 3)\}$. Project 2 requires only an inalienable action from party 2 to match the state of party 3, so $\bar{\mathcal{R}}_2 = \{(2, 3)\}$. All parties possess inalienable means of expropriation ($\mathcal{E} = \varnothing$). Finally, suppose party 1 is more tempted to expropriate party 3's information than party 2, so $\tau^\star_{13} < \tau^\star_{23}$.

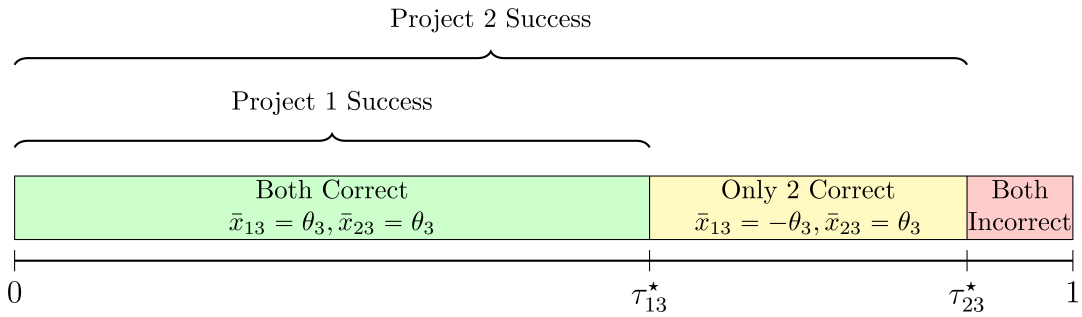

Figure 2: Nested Decision Rule For Example 2

The figure illustrates the realization of recommendations based on the random variable $\xi \sim \text{Unif}[0,1]$. Party 1 receives the correct recommendation only when $\xi \leqslant \tau^\star_{13}$. Party 2 receives the correct recommendation when $\xi \leqslant \tau^\star_{23}$. This decision rule ensures that Party 2 receives the correct recommendation whenever Party 1 receives the correct recommendation.

To deter expropriation, the information conveyed by party i's recommendation for the action $\bar{x}_{i3}$ must be such that $\max\{\tau_{i3}, 1-\tau_{i3}\} \leqslant \tau^\star_{i3}$. Following Example 1, a natural decision rule is to recommend the correct match to party i with probability $\tau^\star_{i3}$ and the incorrect match with probability $1-\tau^\star_{i3}$. However, with two parties required for Project 1, the question is whether these recommendations should be generated independently.

If parties 1 and 2 receive correct recommendations independently, then Project 1 succeeds with probability $\tau^\star_{13} \cdot \tau^\star_{23}$, while Project 2 succeeds with probability $\tau^\star_{23}$. Hence the expected total surplus is $V_1 \cdot \tau^\star_{13} \cdot \tau^\star_{23} + V_2 \cdot \tau^\star_{23}$.

The mediator can do strictly better by *positively correlating* recommendations. Let $\xi \sim \text{Unif}[0,1]$ be a *common* randomization device. Consider the nested rule in which, for each $i \in \{1,2\}$, the mediator privately recommends $\bar{x}_{i3} = \theta_3$ if $\xi \leqslant \tau^\star_{i3}$ and $\bar{x}_{i3} = -\theta_3$ otherwise. This rule is illustrated in Figure 2: Project 1 succeeds with probability $\min\{\tau^\star_{13}, \tau^\star_{23}\} = \tau^\star_{13}$, while Project 2 succeeds with probability $\tau^\star_{23}$. The expected total surplus is

$$V_1 \cdot \min\{\tau^\star_{13}, \tau^\star_{23}\} + V_2 \cdot \tau^\star_{23},$$

which strictly exceeds the independent benchmark. This decision rule ensures that whenever party 1 is recommended the correct action, party 2 is also recommended the correct action.

Proposition 1 below shows that such nested decision rules — implemented via a common randomization device — are optimal in general.

General Results. Fix an owner $g \in \mathcal{N}$. Let $\text{Arm}(g)$ denote the set of "armed" parties, that is parties, who can engage in expropriation under an ownership structure $g \in \mathcal{N}$:

$$\text{Arm}(g) := \{i \in \mathcal{N} : i = g \text{ or } i \notin \mathcal{E}\}.$$

This set includes the owner and all parties with inalienable means for expropriation ($i \notin \mathcal{E}$).

Disarmed parties can be safely fully informed about others because they cannot expropriate. For armed parties, by contrast, posterior beliefs are optimally capped at the expropriation threshold. Let $\tau_{ij}^{\text{safe}}(g)$ denote the maximum precision of party i's belief about θ_j that is consistent with no expropriation:

$$\tau_{ij}^{\text{safe}}(g) := \begin{cases} \tau_{ij}^{\star} = \frac{c_{ij}}{b_{ij}+c_{ij}} & \text{if } i \in \mathrm{Arm}(g) \\ 1 & \text{if } i \notin \mathrm{Arm}(g) \end{cases}.$$

For notational convenience, we set $\tau_{ii}^{\text{safe}}(g) \equiv 1$ for all $i \in \mathcal{N}$ and $g \in \mathcal{N}$.

For project k, it is useful to represent the production requirements as *tasks* of the form "party i must match party j's state." Let

$$\mathcal{Q}_k(g) := \{(g, j) : j \in \mathcal{R}_k\} \ \cup \ \bar{\mathcal{R}}_k \ \subseteq \ \mathcal{N} \times \mathcal{N}$$

denote the set of such tasks that must be performed for project k to succeed under an ownership structure $g \in \mathcal{N}$.

The next proposition establishes that for each project $k \in \mathcal{K}$ the maximum attainable expected surplus is determined by its *bottleneck*: the task with the lowest safety threshold $\tau_{ij}^{\text{safe}}(g)$ among tasks in $(i, j) \in \mathcal{Q}_k(g)$.

Proposition 1 (Optimal Decision Rule)**.** *Fix an ownership structure $g \in \mathcal{N}$. Under Assumptions 1 and 2, the maximum attainable expected total surplus is*

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \left(\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \right). \tag{3.2}$$

Moreover, there exists an optimal decision rule with a nested structure: whenever a task (i, j) with lower safety threshold succeeds, all tasks with higher safety thresholds succeed as well. The mediator draws a single common random variable $\xi \sim \mathrm{Unif}[0, 1]$ and, conditional on (θ, ξ), privately recommends:

(i) No expropriation: each party i is recommended to choose $z_{ij} = \bar{z}_{ij} = 0$ for all $j \neq i$.

(ii) Cooperation: party i is recommended the correct match ($x_{ij} = \bar{x}_{ij} = \theta_j$) if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$, and is recommended the opposite match otherwise ($x_{ij} = \bar{x}_{ij} = -\theta_j$).

Proof. See Section A.1 in the appendix. □

The intuition behind Proposition 1 is as follows. The mediator prefers to avoid expropriation because it destroys total surplus. To do so, information must be partially withheld from *armed* parties (those with the means to expropriate). Therefore, the mediator discloses information to each party i about state θ_j only up to its safety threshold $\tau_{ij}^{\text{safe}}(g)$.

Since tasks are complementary, the probability of successful cooperation in a project is bounded by the bottleneck: the task with the lowest safety threshold, captured by the term $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$. The nested decision rule described in the proposition achieves this bound by correlating agents' recommendations with a common randomization device.

3.2 Optimal Asset Ownership

Proposition 1 characterizes the maximum total surplus $TS(g)$ for each ownership structure $g \in \mathcal{N}$. Optimal asset ownership solves the following problem:

$$g^* \in \arg\max_{g\in\mathcal{N}} TS(g) = \arg\max_{g\in\mathcal{N}} \left\{ \sum_{k\in\mathcal{K}} V_k \cdot \left(\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \right) \right\}.$$

We derive several results on how the optimal asset ownership depends on the model's primitives. To facilitate the discussion, let $\mathcal{P}_k(g)$ denote the set of *active* participants in project k under ownership structure $g \in \mathcal{N}$ — parties who must match *another* party's state for the project to succeed:

$$\mathcal{P}_k(g) := \left\{ i \in \mathcal{N} : \text{there exists some } j \neq i \text{ such that } (i,j) \in \mathcal{Q}_k(g) \right\}. \tag{3.3}$$

Two Roles of Ownership

We start with two examples that clarify the two ways ownership can protect the owner's knowledge from expropriation. First, ownership can confer control over production, in which case the owner's knowledge is protected because it need not be disclosed. Second, ownership can confer control over expropriation, in which case the owner can safely transfer its knowledge because the non-owner lacks the means to expropriate it. The two channels operate through different components of the model: the first affects who participates in production ($\mathcal{P}_k(g)$), while the second affects safety thresholds ($\tau_{ij}^{\text{safe}}(g)$). The following examples illustrate each channel in isolation.

Example 3: Ownership as Control Over Production. Suppose there are two parties, $\mathcal{N} = \{1,2\}$, and a single project, $\mathcal{K} = \{1\}$. The project requires an asset owner to match the states of both parties to succeed, so $\mathcal{R}_1 = \{1,2\}$. No inalienable actions are required ($\bar{\mathcal{R}}_1 = \varnothing$), and all parties possess the inalienable means to expropriate ($\mathcal{E} = \varnothing$).

Given an owner $g \in \{1,2\}$, the project succeeds if and only if $x_{g1} = \theta_1$ and $x_{g2} = \theta_2$. The owner can always match its own state, $x_{gg} = \theta_g$, without disclosing it. By contrast, the non-owner's state must be disclosed to the owner for the project to succeed. Since the owner is armed, the information disclosed about the non-owner's state must be capped to deter expropriation.

Formally, ownership affects the set of active participants, $\mathcal{P}_1(g) = \{g\}$. However, it does not affect safety thresholds ($\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^{\star}$) because expropriation is inalienable ($\mathcal{E} = \varnothing$). The asset should be owned by the party with the weaker temptation to expropriate (equivalently, the higher safety threshold):

$$g^* \in \arg\max_{g \in \mathcal{N}} \left\{ \tau_g^{\star} \cdot V_1 \right\} \quad \Leftrightarrow \quad g^* \in \arg\min_{g \in \mathcal{N}} \left\{ \frac{b_g}{c_g} \right\}, \tag{3.4}$$

where $\tau_g^{\star} := \tau_{gj}^{\star}$, $b_g := b_{gj}$, $c_g := c_{gj}$ for $j \neq g$.

Example 4: Ownership as Control Over Expropriation. Suppose again there are two parties, $\mathcal{N} = \{1, 2\}$, and a single project, $\mathcal{K} = \{1\}$. The project requires an asset owner to match the states of both parties ($\mathcal{R}_1 = \{1, 2\}$). Unlike the previous case, cooperation now requires complementary inalienable actions: each party must use their human capital to match the other party's state ($\bar{\mathcal{R}}_1 = \{(1, 2), (2, 1)\}$). Furthermore, we assume that both parties require the asset to engage in expropriation ($\mathcal{E} = \{1, 2\}$).

There are two key differences from Example 3. First, cooperation now requires inalienable actions by both parties, and therefore ownership does not affect the set of active participants, $\mathcal{P}_1(g) = \{1, 2\}$ for all $g \in \mathcal{N}$. Second, expropriation is alienable and controlled by the owner: the non-owner is disarmed and has no effective expropriation action. Therefore, ownership affects safety thresholds, $\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^{\star}$ if $i = g$ and $\tau_{ij}^{\text{safe}}(g) = 1$ if $i \neq g$.

As in the previous example, the optimal asset owner is the party with the lowest temptation to expropriate (see (3.4)). The logic, however, differs across the two settings. In Example 3, ownership determines who participates in production. In Example 4, ownership determines who lacks the means to expropriate.

Extension. We now highlight the two roles of ownership in a richer setting. Let $\mathcal{Q}_{k,i}(g) := \{j \in \mathcal{N} : (i, j) \in \mathcal{Q}_k(g)\}$ denote the set of targets party i must match for project k to succeed. Let $\tau_{k,i}^{\min}(g) := \min_{j \in \mathcal{Q}_{k,i}(g)} \tau_{ij}^{\text{safe}}(g)$ denote party i's bottleneck in project k (the "hardest" task). For notational convenience, we set $\min \varnothing := 1$.

For the next result, we make the following assumption (in addition to Assumptions 1–2):

Assumption 3. *For every ownership structure $g \in \mathcal{N}$ and project $k \in \mathcal{K}$, non-owner active participants are disarmed: $\mathcal{P}_k(g) \cap \mathrm{Arm}(g) \subseteq \{g\}$.*

Under this assumption, any project in which the owner is *not* an active participant can be implemented with certainty: all of its active participants are disarmed and can safely be fully informed. The only projects that are constrained by an information cap are those in which the owner must act. Thus, the formula for maximum attainable surplus can be simplified.

Corollary 1. *Under Assumption 3, the maximum attainable surplus under an owner $g \in \mathcal{N}$ is*

$$TS(g) = V_{\text{total}} - \sum_{k \in \mathcal{K}} V_k \cdot \mathbf{1}\{g \in \mathcal{P}_k(g)\} \cdot (1 - \tau^{\min}_{k,g}(g)). \tag{3.5}$$

Proof. This follows directly from the formula in (3.2) under Assumption 3 and the definitions of $\mathcal{P}_k(g)$ and $\tau^{\min}_{k,g}(g)$. □

This corollary generalizes the logic of Examples 3 and 4: if there is only one project, and an asset owner is the only armed active participant, then $TS(g) = \tau^\star_g \cdot V_1$, and the asset should be assigned to the party with the lowest temptation to expropriate (highest $\tau^\star_g$). The two terms $\mathbf{1}\{g \in \mathcal{P}_k(g)\}$ and $1 - \tau^{\min}_{k,g}(g)$ in (3.5) reflect production and expropriation channels, respectively.

To isolate the expropriation channel, suppose the owner is an active participant in every project: $g \in \mathcal{P}_k(g)$ for all k. The production channel is constant in g, and the optimal owner is

$$g^* \in \arg\max_{g \in \mathcal{N}} \sum_{k \in \mathcal{K}} V_k \cdot \tau^{\min}_{k,g}(g).$$

Optimal ownership trades off project values against the owner's worst bilateral safety threshold toward the targets it must match.

To isolate the production channel, suppose all parties have the same worst bilateral safety threshold: $\tau^{\min}_{k,g}(g) = \tau^{\min}_k$ for all g. The expropriation channel is constant in g, and the optimal owner becomes

$$g^* \in \arg\min_{g \in \mathcal{N}} \sum_{k \in \mathcal{K}} V_k \cdot \mathbf{1}\{g \in \mathcal{P}_k(g)\} \cdot (1 - \tau^{\min}_k),$$

which depends on g only through the participation indicator. Ownership should not be assigned to a party whose inalienable actions are required in many high-value projects: making such a party the owner would arm it, capping information disclosure across all those projects. Assigning ownership to the *least* involved party allows the more involved parties to be fully informed.

Optimal Ownership and Competition

Next, we study the relationship between competition and optimal ownership. We interpret $\tau^\star_{ij}$ as an inverse measure of competition between i and j: lower $\tau^\star_{ij}$ means that party i can profitably expropriate party j's knowledge at a lower level of posterior precision. The following corollary shows that sufficiently intense competition shifts optimal ownership away from the rival parties.

Corollary 2. *Suppose there is a single project k of value V_k, and competition between i and j is symmetric: $\tau^\star_{ij} = \tau^\star_{ji} = \tau \in (1/2, 1]$. Suppose ownership by either i or j requires the owner to match the other party's state: $(i,j) \in \mathcal{Q}_k(i)$ and $(j,i) \in \mathcal{Q}_k(j)$. Finally, suppose there exists a third party $h \in \mathcal{N}\backslash\{i,j\}$ such that under owner h, neither rival is simultaneously armed and required to match the other rival's state. Let*

$$\bar{\tau} := \min_{(\ell,m)\in\mathcal{Q}_k(h)} \tau^{\mathrm{safe}}_{\ell m}(h).$$

If $\tau < \bar{\tau}$, then every optimal owner satisfies $g^* \in \mathcal{N}\backslash\{i,j\}$.

Proof. Because there is a single project, Proposition 1 implies $TS(g) = V_k \cdot \min_{(\ell,m)\in\mathcal{Q}_k(g)} \tau^{\mathrm{safe}}_{\ell m}(g)$. If $g = i$, then $(i,j) \in \mathcal{Q}_k(i)$ and the owner is armed, so $TS(i) \leqslant \tau V_k$. Similarly, $TS(j) \leqslant \tau V_k$. By assumption, under owner h, neither rival is simultaneously armed and required to match the other rival's state. Hence the bottleneck under h is independent of the rivalry parameter τ, and $TS(h) = \bar{\tau} V_k$. Hence, if $\tau < \bar{\tau}$, then $TS(h) > \max\{TS(i), TS(j)\}$ and $g^* \notin \{i,j\}$. □

This result provides the basis for the knowledge-hub application in Section 4.1.

Optimal Ownership and the Value of Projects

Next, we show that an increase in the value of project $\ell \in \mathcal{K}$ shifts optimal ownership to alleviate the bottleneck in this project. Let $\tau^{\min}_k(g) := \min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\mathrm{safe}}_{ij}(g)$ denote the bottleneck in project $k \in \mathcal{K}$, and let $\Gamma_\ell := \arg\max_g\{\tau^{\min}_\ell(g)\}$ denote the set of owners that maximize it.

Corollary 3. *Holding the values of all projects except ℓ fixed, there exists $\bar{V}_\ell$ such that for $V_\ell > \bar{V}_\ell$, every optimal asset owner belongs to Γ_ℓ.*

Proof. If $\Gamma_\ell = \mathcal{N}$, the result is immediate. Suppose $\Gamma_\ell \neq \mathcal{N}$. Let g^*_ℓ be the owner that maximizes the surplus in projects other than ℓ: $g^*_\ell \in \arg\max_{g'\in\Gamma_\ell} \sum_{k\neq\ell} \tau^{\min}_k(g')V_k$. For any $g \notin \Gamma_\ell$,

$$TS(g^*_\ell) - TS(g) = V_\ell \underbrace{\left[\tau^{\min}_\ell(g^*_\ell) - \tau^{\min}_\ell(g)\right]}_{>0} + \sum_{k\neq\ell} V_k\left[\tau^{\min}_k(g^*_\ell) - \tau^{\min}_k(g)\right].$$

The first term grows linearly in V_ℓ with a strictly positive coefficient, while the second term is independent of V_ℓ. Hence for each $g \notin \Gamma_\ell$ there exists a finite $\bar{V}_\ell(g)$ such that $TS(g^*_\ell) > TS(g)$ when $V_\ell > \bar{V}_\ell(g)$. Taking $\bar{V}_\ell := \max_{g\notin\Gamma_\ell} \bar{V}_\ell(g)$, we have $TS(g^*_\ell) > TS(g)$ for all $g \notin \Gamma_\ell$ when $V_\ell > \bar{V}_\ell$. By construction, $TS(g^*_\ell) \geqslant TS(g')$ for all $g' \in \Gamma_\ell$, so g^*_ℓ is optimal and no owner outside Γ_ℓ can be optimal. □

This result provides the basis for the spin-off application in Section 4.2.

Ownership and Bundling of Control Rights

Finally, we show how the bundling of multiple control rights over production into a single asset prevents achieving the first best.

Corollary 4. *The first best is achieved only when there is at most one alienable requirement:*

$$\max_g TS(g) = V_{\text{total}} \quad \Rightarrow \quad \Big| \bigcup_{k \in \mathcal{K}} \mathcal{R}_k \Big| \leqslant 1.$$

Proof. If there is more than one alienable requirement, then for any ownership assignment $g \in \mathcal{N}$, the asset owner must match the state of at least one party $j \neq g$. Because g is armed, the surplus in projects where $j \in \mathcal{R}_k$ is at most $\tau^\star_{gj} V_k < V_k$, so $TS(g) < V_{\text{total}}$. □

This result provides the basis for the Chinese wall application in Section 4.2.

4 Applications

Our framework highlights that asset ownership may protect knowledge through either control over production or control over expropriation. For example, Apple chose to design the iPad chip in-house rather than partner with Intel because it "just didn't want to teach them everything, which they could go and sell to our competitors" (Steve Jobs, quoted in Isaacson, 2011, p. 493). Here, ownership served to control production and limit knowledge disclosure. In contrast, Apple's ownership choices in its China operations are better interpreted as control over expropriation. Specifically, Apple actively shared knowledge with its Chinese suppliers. Although it did not own most production facilities, it owned key production machines, which limited suppliers' ability to expropriate Apple's know-how (McGee, 2025).

We also show that ownership has costs: non-owners may not be willing to fully disclose their information to the owner. For example, China's quid pro quo policy requires multinationals to form joint ventures with local partners to access the Chinese market. This structure gives the local partner control over some manufacturing decisions, forcing the multinational to disclose production-relevant information. One implication of this policy is that it may discourage automakers from bringing their most advanced technology to China due to concerns about expropriation (Bai et al., 2025).

In this section, we illustrate how our results translate into predictions about an optimal ownership structure and information sharing in several industry environments where two competing downstream firms interact with a common upstream supplier.

Section 4.1 shows how the upstream party can act as a *knowledge hub*. Two competing downstream parties can benefit from pooling knowledge, but they are too tempted to expropriate each other's secrets. A third party that contributes neither private information nor human capital may nonetheless optimally hold the asset solely because its temptation to expropriate is lower. We use this case to rationalize knowledge sharing through third-party suppliers and solution providers, as commonly observed in the auto industry (Küpper et al., 2020).

Section 4.2 studies a different environment in which downstream firms do not need to share knowledge with each other, but each requires an input that only a common upstream supplier can provide. We derive simple conditions under which the supplier must remain independent to maximize total surplus, and when the optimal arrangement instead resembles an internal Chinese wall: the supplier is a subsidiary of one downstream firm, the rival's project succeeds as often as if the supplier were independent, and the parent remains insufficiently informed to expropriate. This application helps explain GM's spin-off of Delphi and provides a rationale for separating Intel's foundry business from its chip-design operations (Fitch, 2025).

4.1 Knowledge Hubs

Consider an extension of Example 3 with three parties, $\mathcal{N} = \{1, 2, 3\}$, and a single project, $\mathcal{K} = \{1\}$. Parties 1 and 2 represent downstream competitors, and party 3 is a common supplier. The project requires the asset owner to match the states of parties 1 and 2 to succeed, so $\mathcal{R}_1 = \{1, 2\}$. No inalienable actions are required, $\bar{\mathcal{R}}_1 = \varnothing$, and all parties possess the inalienable means to expropriate, so $\mathcal{E} = \varnothing$.

In this environment, only the owner's alienable cooperative actions are payoff-relevant. The mediator can therefore keep the two non-owners uninformed, eliminating their expropriation incentives. The remaining constraint is the owner's own temptation: to implement the project, the owner must hold enough information about θ_1 and θ_2 to match them, but not so much as to trigger expropriation. By Proposition 1, total surplus under owner $g \in \{1, 2, 3\}$ is

$$TS(g = 1) = \tau^\star_{12} \cdot V_1, \qquad TS(g = 2) = \tau^\star_{21} \cdot V_1, \qquad TS(g = 3) = \min\{\tau^\star_{31}, \tau^\star_{32}\} \cdot V_1.$$

If the supplier is less tempted than either competitor,

$$\min\{\tau^\star_{31}, \tau^\star_{32}\} \geqslant \max\{\tau^\star_{12}, \tau^\star_{21}\},$$

then it is optimal to assign the asset to the supplier. The distinctive feature of this example is that the optimal owner (party 3) contributes neither private information nor inalienable inputs to production: θ_3 does not enter the cooperative payoffs and $\bar{\mathcal{R}}_1 = \varnothing$. Party 3 is valuable solely because its temptation to expropriate is weaker. In this sense, it functions as a *knowledge hub*, enabling greater information pooling among competing firms.

4.2 Spin-Offs versus Chinese Walls

Suppose there are three parties, $\mathcal{N} = \{1, 2, 3\}$, and two projects, $\mathcal{K} = \{1, 2\}$. We interpret parties 1 and 2 as downstream competitors and party 3 as a common upstream supplier. Project 1 is associated with operations of party 1, and Project 2 is associated with party 2. Both projects require an inalienable input from the supplier: $\bar{\mathcal{R}}_1 = \{(3, 1)\}$ and $\bar{\mathcal{R}}_2 = \{(3, 2)\}$. Project 1 also requires an alienable action by an asset owner: $\mathcal{R}_1 = \{1\}$. For Project 2, we consider two cases: it either requires an alienable action, $\mathcal{R}_2 = \{2\}$, or it does not, $\mathcal{R}_2 = \varnothing$. Expropriation capabilities differ by party: the supplier requires the asset to expropriate, while the downstream parties can expropriate using inalienable means ($\mathcal{E} = \{3\}$).

Cooperative Surplus. Given an asset owner $g \in \{1, 2, 3\}$, the cooperative surplus is

$$V_1 \cdot \mathbf{1}\{\bar{x}_{31} = \theta_1\} \cdot \mathbf{1}\{x_{g1} = \theta_1\} + V_2 \cdot \begin{cases} \mathbf{1}\{\bar{x}_{32} = \theta_2\} \cdot \mathbf{1}\{x_{g2} = \theta_2\} & \text{if} \quad \mathcal{R}_2 = \{2\}, \\ \mathbf{1}\{\bar{x}_{32} = \theta_2\} & \text{if} \quad \mathcal{R}_2 = \varnothing. \end{cases}$$

Bundling of Control Rights. When Project 2 requires the asset ($\mathcal{R}_2 = \{2\}$), the owner has two alienable requirements ($|\mathcal{R}_1 \cup \mathcal{R}_2| = 2$). Thus, by Corollary 4, the first best cannot be achieved. This inextricability lies at the heart of the trade-off between vertical integration and a spin-off. By contrast, when $\mathcal{R}_2 = \varnothing$, the owner is responsible for only one alienable requirement. In this case, the first best can be achieved and resembles an internal Chinese wall.

Subsidiary. If a downstream party owns the asset ($g \in \{1, 2\}$), we call the supplier a subsidiary of that party (vertical integration). If the supplier owns the asset ($g = 3$), we call it an independent upstream firm.

Chinese wall. We say that an *internal Chinese wall* is feasible if there exists an optimal decision rule such that a downstream party's probability of project success when the supplier is a subsidiary of its rival is at least as high as when the supplier is independent. This definition implies that the subsidiary must receive as much information from the downstream non-owner as it would if it were independent. At the same time, the optimality of this rule ensures that the downstream owner (parent) remains insufficiently informed to expropriate.

Case 1: Party 2 Requires the Asset, $\mathcal{R}_2 = \{2\}$. If party 1 owns the asset, then Project 1 can be implemented with certainty: party 1 knows θ_1 and the supplier is disarmed (so θ_1 can be safely disclosed to party 3 to satisfy $\bar{x}_{31} = \theta_1$). Project 2 is different. To make the payoff-relevant alienable decision for Project 2, the owner must match θ_2. Since party 1 is armed, information about θ_2 must be capped at party 1's safety threshold $\tau^{\star}_{12}$. The same logic applies symmetrically if party 2 owns the asset. If instead the supplier owns the asset, then the supplier is armed

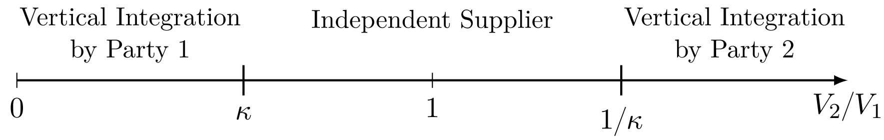

Figure 3: Optimal Ownership When Both Projects Require the Asset

The optimal owner varies with the relative market size V_2/V_1. For this figure, we assume that $\kappa < 1$.

and must receive capped information about both downstream parties' states, so Projects 1 and 2 succeed with probabilities $\tau_{31}^{\star}$ and $\tau_{32}^{\star}$, respectively. Consequently, the maximal attainable surplus under each ownership assignment is

$$TS(g = 1) = V_1 + \tau_{12}^{\star} V_2, \qquad TS(g = 2) = \tau_{21}^{\star} V_1 + V_2, \qquad TS(g = 3) = \tau_{31}^{\star} V_1 + \tau_{32}^{\star} V_2.$$

If the supplier is a subsidiary of party 1 ($g = 1$), party 2's project succeeds with probability $\tau_{12}^{\star}$. If the supplier is independent ($g = 3$), this probability is $\tau_{32}^{\star}$. Because we assume the supplier is less tempted to expropriate ($\tau_{32}^{\star} > \tau_{12}^{\star}$), an internal Chinese wall is not feasible: party 2's project is strictly more likely to succeed under an independent supplier. A symmetric argument applies if party 2 owns the asset.

To simplify the derivation of the optimal ownership structure, suppose that $\tau_D^{\star} := \tau_{12}^{\star} = \tau_{21}^{\star}$ and $\tau_U^{\star} := \tau_{31}^{\star} = \tau_{32}^{\star}$. Recall that the supplier is less tempted to expropriate than a direct competitor, $\tau_U^{\star} > \tau_D^{\star}$. Define the threshold

$$\kappa := \frac{1 - \tau_U^{\star}}{\tau_U^{\star} - \tau_D^{\star}}$$

and assume that $\kappa < 1$. Then vertical integration by party 1 ($g = 1$) is optimal when its project is sufficiently more valuable than party 2's project, $V_2/V_1 < \kappa$. Symmetrically, if $V_2/V_1 > 1/\kappa$, vertical integration by party 2 ($g = 2$) is optimal. For intermediate values, $V_2/V_1 \in [\kappa, 1/\kappa]$, an independent supplier ($g = 3$) is optimal.[1] Figure 3 summarizes the resulting pattern.

Case 2: Party 2 Does Not Require the Asset, $\mathcal{R}_2 = \varnothing$. Relative to Case 1, Project 2 no longer requires the owner to match θ_2. If party 1 owns the asset ($g = 1$), the supplier is disarmed and can be fully informed about both θ_1 and θ_2. Meanwhile, the parent can remain uninformed about θ_2 and match its own state with certainty. Thus, the first-best outcome is achieved.

[1]If $\kappa > 1$, the optimal ownership is $g^* = 1$ if $V_2/V_1 \leqslant 1$ and $g^* = 2$ if $V_2/V_1 > 1$.

The resulting attainable surplus levels for $g \in \{1, 2, 3\}$ are

$$TS(g=1) = V_1 + V_2, \qquad TS(g=2) = \tau^{\star}_{21} V_1 + V_2, \qquad TS(g=3) = \tau^{\star}_{31} V_1 + \tau^{\star}_{32} V_2.$$

When $g = 1$, Project 2 succeeds with probability 1. When $g = 3$, Project 2 succeeds with probability $\tau^{\star}_{32} < 1$. Since $1 > \tau^{\star}_{32}$, an internal Chinese wall is feasible.

Interpretation. Suppose that party 1 is the only downstream firm (incumbent) serving its market (Project 1): $V_1 > 0$ and $V_2 = 0$. Then, it is optimal for this party to own the asset to prevent expropriation by the supplier and safely share knowledge with it. Now assume that a new downstream party enters and serves a different market ($V_2 > 0$). Ideally, they should serve their respective markets, but they can also engage in expropriation. The optimal ownership structure depends on whether the entrant requires the asset that the incumbent party owns to control expropriation by the supplier.

If the entrant does require alienable actions based on this asset, then as the new market V_2 grows in size, the incumbent faces a trade-off: keeping the asset to protect its knowledge versus spinning it off to realize the higher potential of the new market. This interpretation is consistent with GM's spin-off of Delphi. If the entrant does not require actions based on the asset that controls the supplier's expropriation, then it is optimal for the incumbent to keep the asset, and the optimal information structure resembles a Chinese wall between the parent and the subsidiary.

We note that establishing the theoretical possibility of a Chinese wall is distinct from the question of how to build it. Questions regarding the specific internal governance mechanisms and incentive systems required to enforce separation between a parent and its subsidiary lie beyond the scope of this framework and represent an exciting avenue for future research.

5 Discussion

This section clarifies the mechanisms driving our main results by exploring several variations of the baseline model. We establish that the outcome of our information-design approach can implemented in a broader contracting game (Section 5.1), isolate the specific contracting frictions that make ownership matter (Section 5.2), and relax the assumptions of mediated communication (Section 5.3), bundled control rights (Section 5.4), and prohibitive expropriation costs (Section 5.5).

5.1 Implementation Through a Mediated Contract

In the baseline model, we assume that an omniscient mediator can condition recommendations on the true state. Suppose instead that the mediator does not observe the state directly, but parties can use the mediator to exchange information.

Communication Game. Consider the following multi-stage communication game. (i) The state θ is realized, and each party i privately observes θ_i. (ii) Parties confidentially submit reports from some finite message spaces to a neutral mediator, who then implements ownership and transfers and sends private output messages to each party according to a pre-specified mechanism. (iii) Each party privately observes its own output message, together with the implemented ownership and transfers. (iv) Parties simultaneously choose their non-contractible actions. (v) Payoffs are realized.

Mechanism Design. To characterize the set of outcomes that can be supported as a Bayesian Nash equilibrium of some communication game, it is without loss of generality to restrict attention to *direct* mediated mechanisms (Forges, 1986). Specifically, a direct mediated mechanism consists of an allocation rule $\varphi : \Theta \to \Delta(\mathcal{A} \times \mathcal{N})$ and a transfer rule $t : \mathcal{A} \times \mathcal{N} \times \Theta \to \mathcal{T}$. Each party i submits a report $\hat{\theta}_i \in \{-1, 1\}$ to the mediator. Given the report profile $\hat{\theta} = (\hat{\theta}_1, \ldots, \hat{\theta}_N) \in \Theta := \{-1, 1\}^N$, the mediator implements asset ownership $g \in \mathcal{N}$ and sends private action recommendations $a = (a_1, \ldots, a_N) \in \mathcal{A}$ according to $\varphi(a, g \mid \hat{\theta})$. The mediator also enforces monetary transfers according to $t(a, g, \hat{\theta})$. Transfers must be budget balanced, so $\mathcal{T}$ is the set of budget-balanced transfer vectors.

An outcome is a joint distribution over actions, ownership, transfers, and states. Feasible outcomes must satisfy incentive-compatibility constraints: no party should gain by misreporting its state, disobeying its recommendation, or combining the two deviations. Our first result shows that the maximal total surplus in this communication game coincides with the value of the information-design problem.

Proposition 2. *The maximum attainable surplus in the communication game is* $\max_{g \in \mathcal{N}} TS(g)$*. This value is attained by a direct mediated mechanism.*

The intuition behind this proposition is as follows. For fixed ownership, the set of feasible outcomes is characterized by obedience, truth-telling, and no-double-deviation constraints. Therefore, the maximum attainable surplus cannot exceed the value of the benchmark mediator problem, which imposes only obedience. Randomizing over ownership does not help relax these constraints because parties observe ownership when choosing their non-contractible actions (a party that operates the asset knows that it operates the asset).

The allocation rule that achieves the maximum possible surplus is given by

$$\varphi^*(a, g^* \mid \theta) = \sigma^*(a \mid \theta) \qquad \text{and} \qquad \varphi^*(a, g \mid \theta) = 0 \text{ for } g \neq g^*, \tag{5.1}$$

where $g^* \in \arg\max_g TS(g)$ and $\sigma^*(a \mid \theta)$ is the optimal decision rule under g^*. We show that, under constant transfers, this mechanism is incentive compatible. Intuitively, once no armed party receives enough information to profitably expropriate, the remaining incentives are aligned, so truthful reporting can be sustained. Thus, the central friction in our setting is the non-contractibility of actions rather than private information per se.

Contracting Game. We now embed the communication game in a broader bargaining environment. Fix an initial owner $g_0 \in \mathcal{N}$. A *contract* is a direct mediated mechanism (φ, t), where φ is an allocation rule and t is a budget-balanced transfer rule, as defined above.

The contracting game has two additional stages before the communication game. First, party 1 makes a take-it-or-leave-it offer of a contract (φ, t) to the other parties. Second, each party $i \in \{2, \ldots, N\}$ either accepts or rejects. If all parties accept, the communication game induced by (φ, t) is played. If at least one party rejects, the contract is void, the asset owner remains g_0, transfers are zero, no information is shared, and parties choose actions in the disagreement environment.

In Appendix A.2, we show that there exists a Perfect Bayesian Equilibrium in which party 1 offers the direct mediated contract (φ^*, t^*), where φ^* is given by (5.1) and t^* ensures participation. All other parties accept, and, conditional on acceptance, parties report truthfully and obey their recommendations.

5.2 Contractibility of Actions

In the baseline model, we assume that all actions are non-contractible. If some actions are contractible, then whenever the mediator recommends those actions they are enforced, which relaxes the obedience constraints. Thus, contractibility weakly increases the maximum attainable total surplus. This section outlines the results; formal proofs are provided in Appendix A.3.

Expropriation. If *expropriation* actions $(z, \bar{z})$ are contractible, then the mediator can directly enforce $z_{ij} = \bar{z}_{ij} = 0$ before any information is disclosed. Once expropriation is ruled out, the mediator can fully reveal information and achieve the first best under *any* ownership structure.

We also consider *partial* contractibility of expropriation. Here the expropriation action itself remains non-contractible, but a noisy signal of expropriation is verifiable. Suppose that after actions are chosen, a signal $s^e_{ij} \in \{0, 1\}$ indicates an expropriation attempt by party i against party j. If $\zeta_{ij} \neq 0$, then $s^e_{ij} = 1$ with probability $\rho^e_{ij} \in [0,1]$; if $\zeta_{ij} = 0$, then $s^e_{ij} = 0$ surely. A contract may specify a transfer $t^e_{ij} \leqslant \bar{t}_{ij}$ from i to j when $s^e_{ij} = 1$. The optimal penalty is $t^e_{ij} = \bar{t}_{ij}$, which raises the safety threshold to

$$\tilde{\tau}^{\text{safe}}_{ij}(g) := \begin{cases} \min\left\{1, \dfrac{c_{ij} + \rho^e_{ij}\bar{t}_{ij}}{b_{ij} + c_{ij}}\right\} & \text{if } i \in \text{Arm}(g), \\ 1 & \text{if } i \notin \text{Arm}(g). \end{cases}$$

Thus partial contractibility continuously raises the safety threshold, interpolating between the non-contractible case ($\rho^e_{ij}\bar{t}_{ij} = 0$) and the fully contractible case ($\rho^e_{ij}\bar{t}_{ij} \geqslant b_{ij}$).

Cooperation. If *cooperative* actions $(x, \bar{x})$ and project shares (α_{ik}) are contractible while expropriation remains non-contractible, then the fundamental constraint is unchanged: after learning information, an armed party can still deviate toward expropriation. Therefore, information disclosure must still satisfy the same *expropriation obedience* constraints as in the baseline model. As a result, the maximal attainable total surplus is governed by the same obedience-based bound as in the baseline problem (i.e., it cannot exceed $\max_g TS(g)$).

If only the cooperative shares (α_{ik}) are contractible (without making $(x, \bar{x})$ contractible), this cannot increase attainable total surplus either. The "shares-only" environment is a restriction of the "contractible cooperation and shares" environment, so its value is weakly smaller. Since the latter yields no improvement over the baseline and the baseline policy remains feasible when shares are contractible, all three values coincide.

5.3 Non-Contractible Communication: Cheap Talk

In the baseline setting, the mediator plays two distinct roles: it garbles information to deter expropriation (limiting the posterior precision of armed receivers), and it coordinates disclosure by correlating recommendations across parties. In this subsection, we examine the role of the omniscient mediator by contrasting it with unmediated *cheap talk*, where each sender can condition its message only on its own state.

We show that with a sufficiently rich state and action space, direct cheap talk can replicate the mediator's information-garbling role without any need for commitment to randomization. Thus, in environments that require only information garbling, such as the spin-off or Chinese wall applications, the results remain the same under cheap-talk communication. However, the mediator's coordination role generally cannot be replicated. Because parties send messages independently rather than through a central coordinator, total surplus under the O-ring technology is reduced by uncoordinated disclosure.

Cheap Talk with Binary State. Suppose party j must disclose its information to exactly one armed party i. To deter expropriation, i's posterior precision regarding θ_j must not exceed the threshold $\tau^\star_{ij} \in (\frac{1}{2}, 1)$. Under the optimal mediated decision rule, the mediator ensures this constraint binds by committing to a noisy disclosure: it recommends the correct action with probability $\tau^\star_{ij}$ and the incorrect action with probability $1 - \tau^\star_{ij}$. This rule maximizes the probability of successfully completing task (i, j) while deterring expropriation.

Under direct communication, party j lacks the ex-ante commitment power to implement this garbling. Any informative pure strategy profile must be fully revealing, in which case the armed receiver i would expropriate the information, giving sender j a profitable deviation. A babbling equilibrium always exists, but it yields a strictly lower probability of task success than the mediated benchmark.

To attain the mediator's success probability $\tau^\star_{ij}$, the sender must therefore randomize across messages. However, because j lacks commitment, it must evaluate its messages ex-post (after observing the true state). Party j is willing to randomize its disclosure only if it is indifferent between reporting the true state and lying. For this indifference condition to hold, the receiver i must also play a mixed strategy, randomizing between attempting expropriation and no expropriation to keep the sender indifferent. Thus, any cheap-talk equilibrium that matches the mediator's probability of cooperative success must involve strictly positive expropriation on the equilibrium path, decreasing total expected surplus.

Rich State Space: Continuum Analogue. The failure of unmediated cheap talk to achieve optimal garbling is an artifact of the binary state space. With a binary state, any *pure* reporting strategy induces a posterior belief of either $\frac{1}{2}$ or 1. Therefore, reaching the indifference threshold $\tau^\star_{ij} \in (\frac{1}{2}, 1)$ requires the sender to randomize. However, by enriching the state space to a continuum, the sender can induce any posterior between $\frac{1}{2}$ and 1 through *pure* strategies. Thus, the lack of commitment to randomize is not a fundamental economic friction in our environment.

Formally, suppose instead of a binary state each party i privately observes an independent state θ_i uniformly distributed on a unit circle $\mathbb{T} = [0, 1)$. Cooperative actions x_{ij} and $\bar{x}_{ij}$ are chosen from $\mathbb{T}$. Expropriation actions z_{ij} and $\bar{z}_{ij}$ are chosen from $\mathbb{T} \cup \{\varnothing\}$, where $\varnothing$ denotes the safe action of not attempting expropriation.

Actions succeed if they fall within a target-specific circular tolerance $\delta_j > 0$ of the true state, using the circular distance $d(x, y) := \min\{|x - y|,\ 1 - |x - y|\}$. Specifically, for an owner $g \in \mathcal{N}$, an alienable task $j \in \mathcal{R}_k$ succeeds if $d(x_{gj}, \theta_j) \leqslant \delta_j$, and an inalienable task $(i, j) \in \bar{\mathcal{R}}_k$ succeeds if $d(\bar{x}_{ij}, \theta_j) \leqslant \delta_j$. As in the baseline model, party i's effective expropriation action is ζ_{ij}. Expropriation succeeds if $\zeta_{ij} \in \mathbb{T}$ and $d(\zeta_{ij}, \theta_j) \leqslant \delta_j$, yielding the same payoffs for successful and unsuccessful attempts as the baseline model.

We impose the following continuous analogue of Assumption 2:

Assumption 2*. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $b_{ij} < c_{ij}$ and $\delta_j \in (0, 1/4]$.*

This assumption ensures that an uninformed party has no incentive to expropriate. Specifically, under the uniform prior, if an uninformed party i guesses the state θ_j, it succeeds with probability $2\delta_j$. Because the benefit of successful expropriation is strictly lower than the cost of an unsuccessful attempt ($b_{ij} < c_{ij}$), it follows that the expropriation threshold is $\tau^\star_{ij} = \frac{c_{ij}}{b_{ij}+c_{ij}} > 2\delta_j$. Therefore, without additional information, the party will not expropriate.

Cheap Talk with Continuous State. Consider the cheap talk under a continuous state and action space. For simplicity, we restrict attention to environments in which each sender's state must be communicated to at most one *armed* party. Formally, fix an owner g and suppose that for every target $j \in \mathcal{N}$, there is at most one party $i \in \text{Arm}(g)$ such that $(i, j) \in \bigcup_{k \in \mathcal{K}} \mathcal{Q}_k(g)$. This assumption covers Examples 3 and 4 and all applications from Section 4. We prove the following result.

Proposition 3 (Direct Garbling)**.** *Consider the continuum analogue of our environment. Suppose that each target state must be communicated to exactly one armed party. Under Assumptions 1 and 2*, there exists a pure-strategy Perfect Bayesian Equilibrium such that every task (i, j) succeeds with probability $\tau^{\text{safe}}_{ij}(g)$, and there is no expropriation on the equilibrium path.*

Proof. See Appendix A.4 □

The PBE is constructed via deterministic pooling of states, as illustrated in Figure 4. For each task (i, j) with an armed receiver i, the state space is partitioned into a success region $[0, \tau^\star_{ij}]$ and a failure region $(\tau^\star_{ij}, 1)$. Both regions are divided into M intervals $\{S^{(m)}\}_{m=1}^M$ and $\{F^{(m)}\}_{m=1}^M$ such that $\bigcup_{m=1}^M S^{(m)} = [0, \tau^\star_{ij}]$ and $\bigcup_{m=1}^M F^{(m)} = (\tau^\star_{ij}, 1)$, where $M = \lceil \frac{\tau^\star_{ij}}{2\delta_j} \rceil$ is the number of messages j can send to i.

In equilibrium, party j sends message $m \in \{1, 2 \ldots, M\}$ to i if $\theta_j \in S^{(m)} \cup F^{(m)}$. The lengths of the success and failure fragments are proportional such that

$$\frac{|S^{(m)}|}{|S^{(m)}| + |F^{(m)}|} = \tau^\star_{ij}.$$

Under this partition, the armed receiver i is indifferent between expropriating and not expropriating, and optimally selects the intended cooperative action $x^{(m)}$ that covers the success region $S^{(m)}$. Crucially, the sender j has no incentive to deviate: if $\theta_j \in S^{(m)}$, then sending m ensures the task (i, j) succeeds; if $\theta_j \in F^{(m)}$, the task fails regardless of the message sent, making the sender indifferent across all messages.

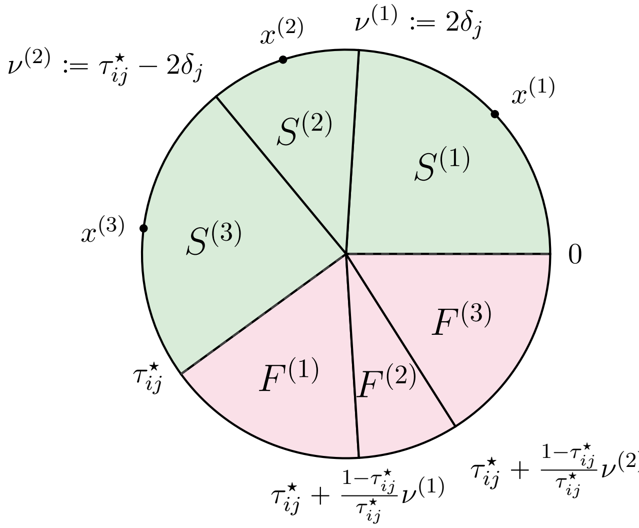

Figure 4: State space partition for the cheap-talk PBE

Party j sends message $m \in \{1, \ldots, M\}$ to armed party i if $\theta_j \in S^{(m)} \cup F^{(m)}$. This example is constructed under $\delta_j = 0.12$ and $\tau^\star_{ij} = 0.6$, where $M = \lceil \frac{\tau^\star_{ij}}{2\delta_j} \rceil = 3$. The cooperative actions corresponding to message $m \in \{1, 2, 3\}$ are $x^{(1)} = \delta_j$, $x^{(2)} = \tau^\star_{ij}/2$, and $x^{(3)} = \tau^\star_{ij} - \delta_j$.

The Mediator's Benchmark. In the continuum analogue, the omniscient mediator problem has the same bottleneck logic as the baseline binary model (see Appendix A.4). To rule out expropriation, each task (i, j) cannot succeed with a probability higher than its safety threshold $\tau^{\text{safe}}_{ij}(g)$. Therefore, project k cannot succeed with a probability higher than the minimum of these safety thresholds across all required tasks.

We construct an obedient decision rule that achieves the upper bound implied by the safety thresholds, delivering the same total surplus formula as in Proposition 1 (see Equation 3.2). Consider the continuum adaptation of the optimal decision rule in the binary case. Define the *safety arc length* as

$$L_{ij}(g) := \frac{2\delta_j}{\tau^{\text{safe}}_{ij}(g)},$$

and let $\xi \sim \text{Unif}[0, 1]$ be a common randomization device. For self-tasks, the mediator recommends $x_{ii} = \bar{x}_{ii} = \theta_i$. For non-self tasks $j \neq i$, the mediator recommends

$$x_{ij} = \bar{x}_{ij} = \theta_j + L_{ij}(g)\left(\frac{1}{2} - \xi\right) \pmod 1, \qquad z_{ij} = \bar{z}_{ij} = \varnothing.$$

Geometrically, the mediator creates an arc of length $L_{ij}(g)$ centered at the true state θ_j, and then draws a recommendation uniformly at random from this arc. Conditional on receiving this recommendation, the posterior of an armed party is uniform over an interval of length $L_{ij}(g)$ with the recommended action as its midpoint. Therefore, the maximal probability of success for either the productive task or expropriation is $2\delta_j/L_{ij}(g) = \tau_{ij}^{\text{safe}}(g)$, which deters expropriation.

Proposition 3 implies that the cheap-talk environment delivers the same total surplus as the mediated benchmark in Examples 3 and 4, as well as in the spin-off and Chinese-Wall applications. Because each project in those environments relies on at most one task with an armed receiver, the optimal information disclosure requires only information garbling, which can be replicated through direct communication with a rich enough state and action space.

The knowledge-hub application, however, is different. When the supplier acts as the hub ($g = 3$), it must learn both θ_1 and θ_2 from parties 1 and 2 to implement the project. Proposition 3 ensures that both tasks $(3, 1)$ and $(3, 2)$ can be independently implemented via cheap talk with success probabilities $\tau_{31}^{\star}$ and $\tau_{32}^{\star}$. However, because parties can condition their messages only on their own private states, the disclosures are independent. This lack of coordination yields a strictly lower expected surplus:

$$TS^{\text{cheap talk}}(g = 3) = \tau_{31}^{\star} \cdot \tau_{32}^{\star} \cdot V_1 < \min\{\tau_{31}^{\star}, \tau_{32}^{\star}\} \cdot V_1 = TS(g = 3).$$

Thus, unmediated communication cannot replicate centralized conditioning of recommendations on the full state vector. Moreover, in the knowledge-hub application, such conditioning does not even require exogenous randomization by the mediator. In Appendix A.4, we show that instead of drawing $\xi \sim \text{Unif}[0, 1]$, the mediator can use one sender's state to coordinate disclosure from both senders to the hub. The maximum total surplus can therefore be attained by a *deterministic* obedient decision rule.

5.4 Multiple Assets and Separation of Control Rights

In the baseline model, a single asset bundles multiple alienable decision rights. This bundling generates the central trade-off: ownership protects the owner's knowledge but leaves others exposed to expropriation, discouraging disclosure. This subsection extends the model to multiple assets and allows control rights to be unbundled.

Assets. Let $\mathcal{M}$ be a finite set of assets. Two assignment functions determine which asset controls each *alienable* decision:

$$\phi^c : \mathcal{N} \to \mathcal{M}, \qquad \phi^e : \mathcal{N} \to \mathcal{M}.$$

Here $\phi^c(j)$ is the asset whose owner controls the alienable *cooperative* action targeting state θ_j, while $\phi^e(j)$ is the asset whose owner controls the alienable *expropriation* action targeting θ_j (for asset-dependent expropriators). The baseline model with one asset corresponds to $|\mathcal{M}| = 1$, so a single party controls *all* alienable actions. In general, control over production decisions is bundled when $\phi^c(i) = \phi^c(j)$ for $i \neq j$; control over production and expropriation is bundled when $\phi^c(j) = \phi^e(j)$.

An ownership profile is $g = (g_i)_{i \in \mathcal{M}} \in \mathcal{N}^{\mathcal{M}}$. For each target j, define the induced owners

$$g^c(j) := g_{\phi^c(j)}, \qquad g^e(j) := g_{\phi^e(j)}.$$

Production. Project k succeeds if and only if the relevant alienable and inalienable matching requirements are satisfied: $x_{g^c(j)\,j} = \theta_j$ for all $j \in \mathcal{R}_k$, and $\bar{x}_{\ell j} = \theta_j$ for all $(\ell, j) \in \bar{\mathcal{R}}_k$. The set of matching tasks for project k under ownership profile g is

$$\mathcal{Q}_k(g) := \{(g^c(j), j) : j \in \mathcal{R}_k\} \;\cup\; \bar{\mathcal{R}}_k. \tag{5.2}$$

Expropriation. A party i can expropriate target $j \neq i$ either via inalienable means ($i \notin \mathcal{E}$) or via an asset if $i \in \mathcal{E}$ and $i = g^e(j)$. Thus the set of parties who are armed *against target* j is

$$\mathrm{Arm}_j(g) := \{i \in \mathcal{N} : i = g^e(j) \text{ or } i \notin \mathcal{E}\}.$$

Define the target-specific safety threshold (and set $\tau_{ii}^{\mathrm{safe}}(g) := 1$)

$$\tau_{ij}^{\mathrm{safe}}(g) \;:=\; \begin{cases} \tau_{ij}^{\star} = \dfrac{c_{ij}}{b_{ij} + c_{ij}} & \text{if } i \in \mathrm{Arm}_j(g) \text{ and } i \neq j, \\ 1 & \text{otherwise.} \end{cases} \tag{5.3}$$

Total Surplus. Under the same parameter restrictions as in Proposition 1 (Assumptions 1–2), the characterization of the maximum attainable total surplus carries over:

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g),$$

where $\mathcal{Q}_k(g)$ and $\tau_{ij}^{\mathrm{safe}}(g)$ are defined in (5.2) and (5.3), respectively.

Since the baseline model corresponds to $|\mathcal{M}| = 1$ and $\phi^c := \phi^e$, allowing separable control rights weakly increases maximum attainable total surplus. The following proposition shows how unbundling control rights across assets can achieve the first best.

Proposition 4 (First best under separable control rights)**.** *The first best V_{total} is achievable in each of the following cases.*

(i) All cooperative requirements are alienable ($\bar{\mathcal{R}}_k = \varnothing$ for all k), and distinct targets are controlled by distinct cooperative assets (ϕ^c is injective).

(ii) All expropriation is asset-dependent ($\mathcal{E} = \mathcal{N}$), and distinct targets are governed by distinct expropriation assets (ϕ^e is injective).

Proof. (i) Because ϕ^c is injective, we can choose an ownership profile g such that $g^c(j) = j$ for all $j \in \mathcal{N}$. Since $\bar{\mathcal{R}}_k = \varnothing$, each project has task set $\mathcal{Q}_k(g) = \{(j, j) : j \in \mathcal{R}_k\}$, which is trivial because party j observes θ_j. Hence $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\text{safe}}_{ij}(g) = 1$ for every k and therefore $TS(g) = V_{\text{total}}$.

(ii) Because ϕ^e is injective, choose g such that $g^e(j) = j$ for all $j \in \mathcal{N}$. Fix any $i \neq j$. Since $\mathcal{E} = \mathcal{N}$, party i has no inalienable expropriation; and because $g^e(j) = j \neq i$, party i also does not control the expropriation asset for target j. Hence $i \notin \text{Arm}_j(g)$ and $\tau^{\text{safe}}_{ij}(g) = 1$ for all $i \neq j$. It follows that for every project k, $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\text{safe}}_{ij}(g) = 1$, so again $TS(g) = V_{\text{total}}$. □

The intuition behind Proposition 4 is as follows. When cooperation requires only alienable actions and ownership can be arranged so that each action that depends on party i's information is controlled by party i, then no private information needs to be disclosed. Parties can therefore avoid expropriation because they can remain silent while still implementing the required cooperative decisions. This case shows how unbundling production decision rights can achieve the first best in Example 3 (ownership as control over production).

When expropriation requires access to assets and ownership can be arranged so that each party controls the asset needed to expropriate *its own* information, then parties can block all expropriation attempts. In that case, the first best is achievable because the means of expropriation are held by the original knowledge holder rather than the potential expropriator. This case shows how unbundling production and expropriation decision rights can achieve the first best in Example 4 (ownership as control over expropriation).

5.5 Optimal Decision Rule Under Arbitrary Expropriation Costs

The baseline model assumes successful expropriation is prohibitively costly, $\lambda_{ij} - b_{ij} > V_{\text{total}}$, so the mediator always prefers to deter expropriation. When expropriation is costly but not prohibitive, it may be optimal to tolerate some expropriation in order to raise the probability of success. The optimal ownership structure may then depend on the absolute cost of successful expropriation λ_{ij}, not just on the temptation to expropriate $\tau^\star_{ij} := \frac{c_{ij}}{b_{ij}+c_{ij}}$.

To build intuition, consider the extension of Example 3 under arbitrary expropriation costs: two parties $\mathcal{N} = \{1, 2\}$ and a single project $\mathcal{K} = \{1\}$; the project requires only alienable actions ($\mathcal{R}_1 = \{1, 2\}$, $\bar{\mathcal{R}}_1 = \varnothing$), and each party can expropriate ($\mathcal{E} = \varnothing$). Since the non-owner does not participate in production, only the owner must learn the non-owner's state.

The optimal decision rule takes one of two forms (Figure 5): either partial disclosure at the safety thresholds when expropriation costs are high or full disclosure when expropriation costs

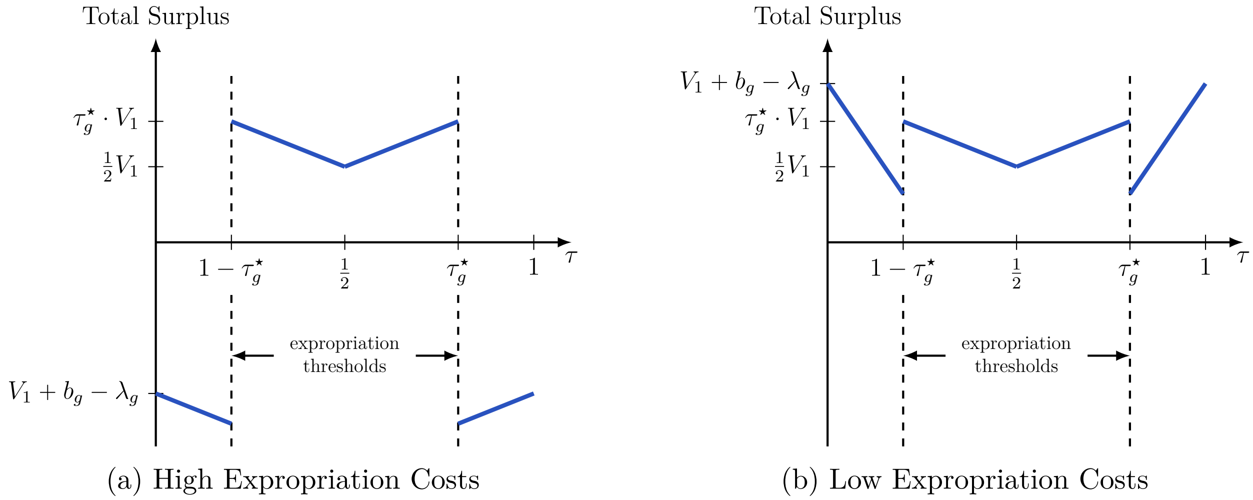

Figure 5: Optimal Disclosure in Example 3 Under Low and High Expropriation Costs

Both panels show total expected surplus as a function of the owner's belief that the non-owner's state is $\theta_j = 1$.

are low. The maximum total surplus under owner g is therefore

$$TS(g) = \max\{\tau_g^\star \cdot V_1,\ V_1 - (\lambda_g - b_g)\},$$

where $\tau_g^\star := \tau_{gj}^\star$ and $b_g := b_{gj}$ for the non-owner $j \neq g$.

In the baseline model, $\lambda_g - b_g$ is large for both $g \in \{1, 2\}$, so the first term always dominates. The optimal owner is then the party with the lower temptation to expropriate (higher $\tau_g^\star$). Under arbitrary costs, the second term can dominate. Suppose party 1 has a higher temptation to expropriate ($\tau_1^\star < \tau_2^\star$) but cheaper expropriation ($\lambda_1 - b_1 < V_1 < \lambda_2 - b_2$). If $\lambda_1 - b_1$ is sufficiently close to zero, assigning the asset to party 1 becomes optimal: full disclosure achieves near-first-best surplus at a small expropriation cost, outperforming the safety-threshold solution available under party 2's ownership.

Appendix A.5 extends this characterization to settings in which only one armed party must receive information about others, covering all applications in the paper.

6 Conclusion

This paper studies how knowledge is allocated across parties when it can be used both productively and opportunistically. In the absence of contractual tools to prevent opportunistic behavior, ownership determines who must be informed and who has the means to expropriate. However, reallocating ownership is an imperfect substitute for complete contracts: it moves a coarse bundle of decision rights rather than the targeted control needed to achieve efficient cooperation without expropriation risk. This framework helps explain a number of industry practices, such as organizing knowledge pooling through common partners, spinning off suppliers to facilitate greater knowledge disclosure from rivals, and the theoretical possibility of Chinese Walls within organizations.

This framework also suggests a few directions for future work. First, our model takes a step toward studying how transactions across firms are shaped by their internal structures. In particular, we provide a simple condition on the production technology that makes internal Chinese walls within organizations impossible. However, even if such walls are theoretically possible, their credibility still depends on governance and internal incentives. Incorporating richer organizational structure into this framework is a natural next step.

Second, in our mechanism-design formulation, parties commit to a mediated contract before observing their private information. If contracting is instead initiated by an informed party, the contract offer itself might convey information. In standard economic models, private information is usually represented as the realization of a random variable observed by some parties but not others, drawn from a common-knowledge distribution. In such settings, information can in principle be concealed through the design of the mechanism — an insight known as the *inscrutability principle* (Myerson, 1983).

The inscrutability logic can fail when information is "eye-opening," in the sense that it changes how parties perceive the relevant state space or feasible actions (Tirole, 2009). Then the mere act of specifying previously unforeseen contingencies expands the other party's awareness and shifts beliefs. Even if parties can contractually rule out expropriation, the disclosure required to initiate contracting changes outside options relative to no contracting, so the initial ownership structure can shape knowledge flows even when expropriation is contractible. Formalizing this setting likely requires an alternative model of information (e.g., Akerlof, Holden, and Li, 2025) and is a promising direction for future research.

References

Akerlof, R., Holden, R., & Li, H. (2025). Getting the Picture. *Working Paper*.

Alonso, R., Dessein, W., & Matouschek, N. (2008). When Does Coordination Require Centralization? *American Economic Review*, *98*(1), 145–179.

Anton, J. J., & Yao, D. A. (1994). Expropriation and Inventions: Appropriable Rents in the Absence of Property Rights. *The American Economic Review*, *84*(1), 190–209.

Anton, J. J., & Yao, D. A. (2002). The Sale of Ideas: Strategic Disclosure, Property Rights, and Contracting. *The Review of Economic Studies*, *69*(3), 513–531.

Arrow, K. (1962). Economic Welfare and the Allocation of Resources for Invention. In R. Nelson (Ed.), *The Rate and Direction of Incentive Activity: Economic and Social Factors* (pp. 609–626). Princeton University Press.

Atalay, E., Hortaçsu, A., & Syverson, C. (2014). Vertical Integration and Input Flows. *American Economic Review*, *104*(4), 1120–1148.

Bai, J., Barwick, P. J., Cao, S., & Li, S. (2025). Quid Pro Quo, Knowledge Spillovers, and Industrial Quality Upgrading: Evidence from the Chinese Auto Industry. *American Economic Review*, *115*(11), 3825–3852.

Baliga, S., & Sjöström, T. (2018). A Theory of the Firm Based on Haggling, Coordination, and Rent-Seeking.

Bergemann, D., & Morris, S. (2019). Information Design: A Unified Perspective. *Journal of Economic Literature*, *57*(1), 44–95.

Bradsher, K. (1998). G.M. Plans to Spin Off Parts Division. *The New York Times*.

Demirer, M., & Karaduman, Ö. (2025). Do Mergers and Acquisitions Improve Efficiency? Evidence from Power Plants. *Working Paper*.

Dessein, W. (2002). Authority and Communication in Organizations. *Review of Economic Studies*, *69*(4), 811–838.

Fitch, A. (2025). Is Nvidia Intel's Savior? Not Quite. *The Wall Street Journal*.

Forges, F. (1986). An Approach to Communication Equilibria. *Econometrica*, *54*(6), 1375–1385.

Fudenberg, D., & Rayo, L. (2019). Training and Effort Dynamics in Apprenticeship. *American Economic Review*, *109*(11), 3780–3812.

Garicano, L., & Rayo, L. (2017). Relational Knowledge Transfers. *American Economic Review*, *107*(9), 2695–2730.

Grossman, S., & Hart, O. (1986). The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration. *Journal of Political Economy*, *94*(4), 691–719.

Hart, O. (1995). *Firms, Contracts, and Financial Structure (Clarendon Lectures in Economics)*. Oxford University Press.

Hart, O., & Moore, J. (1990). Property Rights and the Nature of the Firm. *Journal of Political Economy*, *98*(6), 1119–1158.

Helper, S., & Henderson, R. (2014). Management Practices, Relational Contracts, and the Decline of General Motors. *Journal of Economic Perspectives*, *28*(1), 49–72.

Hörner, J., & Skrzypacz, A. (2016). Selling Information. *Journal of Political Economy*, *124*(6), 1515–1562.

Isaacson, W. (2011). *Steve Jobs*. Simon & Schuster.

Kamenica, E., & Gentzkow, M. (2011). Bayesian Persuasion. *American Economic Review*, *101*(6), 2590–2615.

Klemperer, P., Cramton, P., & Gibbons, R. (1987). Dissolving a Partnership Efficiently. *Econometrica*, *55*(3), 615–632.

Kolotilin, A., & Li, H. (2021). Relational Communication. *Theoretical Economics*, *16*(4), 1391–1430.

Kremer, M. (1993). The O-Ring Theory of Economic Development. *The Quarterly Journal of Economics*, *108*, 551–575.

Kreps, D. M., & Wilson, R. (1982). Sequential Equilibria. *Econometrica*, *50*(4), 863–894.

Küpper, D., Okur, A., Betti, F., Bezamat, F., Fendri, M., & Fernandez, B. (2020). How Manufacturers Can Unlock Value from Data Sharing. *Boston Consulting Group*.

Liebeskind, J. (1996). Knowledge, Strategy, and the Theory of the Firm. *Strategic Management Journal*, *17*, 93–107.

Marshall, R. C., & Marx, L. M. (2007). Bidder Collusion. *Journal of Economic Theory*, *133*(1), 374–402.

Mathevet, L., Perego, J., & Taneva, I. (2020). On Information Design in Games. *Journal of Political Economy*, *128*(4), 1370–1404.

Matouschek, N. (2004). Ex Post Inefficiencies in a Property Rights Theory of the Firm. *The Journal of Law, Economics, and Organization*, *20*(1), 125–147.

McGee, P. (2025). *Apple in China: The Capture of the World's Greatest Company*. Simon; Schuster.

Myerson, R. (1983). Mechanism Design by an Informed Principal. *Econometrica*, *51*(6), 1767–1797.

Ortner, J., Sugaya, T., & Wolitzky, A. (2024). Mediated Collusion. *Journal of Political Economy*, *132*(4), 1247–1289.

Rantakari, H. (2008). Governing Adaptation. *Review of Economic Studies*, *75*(4), 1257–1285.

Rayo, L., & Segal, I. (2010). Optimal Information Disclosure. *Journal of Political Economy*, *118*(5), 949–987.

Segal, I., & Whinston, M. D. (2016). Property Rights and the Efficiency of Bargaining. *Journal of the European Economic Association*, *14*(6), 1287–1318.

Sugaya, T., & Wolitzky, A. (2018). Maintaining Privacy in Cartels. *Journal of Political Economy*, *126*(6), 2569–2607.

Teece, D. (1980). Economies of Scope, and the Scope of the Enterprise. *Journal of Economic Behavior and Organization*, *1*(3), 223–247.

Tirole, J. (2009). Cognition and Incomplete Contracts. *American Economic Review*, *99*(1), 265–294.

A Appendix

A.1 Proof of Proposition 1

Recall that $\mathcal{R}_k$ and $\bar{\mathcal{R}}_k$ are alienable and inalienable requirements, respectively, and $\mathcal{Q}_k(g) := \{(g,j) : j \in \mathcal{R}_k\} \cup \bar{\mathcal{R}}_k \subseteq \mathcal{N} \times \mathcal{N}$ is the set of production requirements under owner $g \in \mathcal{N}$. Proposition 1 states that the maximum attainable expected total surplus under owner $g \in \mathcal{N}$ is

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \left(\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \right). \tag{A.1}$$

Moreover, there exists an optimal obedient decision rule that is *nested*. Let $\xi \sim \text{Unif}[0,1]$ be a common randomization device. For each (ξ, θ, g) define an action profile $a(\xi, \theta, g) = (x, \bar{x}, z, \bar{z})$ as follows:

$$z_{ij} = 0 \ \text{ and } \ \bar{z}_{ij} = 0 \qquad \text{for all } i \in \mathcal{N},\ j \neq i, \tag{A.2}$$

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j & \text{if } \xi \leqslant \tau_{ij}^{\text{safe}}(g), \\ -\theta_j & \text{if } \xi > \tau_{ij}^{\text{safe}}(g), \end{cases} \qquad \text{for all } i, j \in \mathcal{N}. \tag{A.3}$$

Define $\sigma^g(\cdot \mid \theta) \in \Delta(\mathcal{A})$ as the distribution induced by $\xi \sim \text{Unif}[0,1]$ and $a = a(\xi, \theta; g)$; that is,

$$\sigma^g(a \mid \theta) = \mathbb{P}_\xi\big[a(\xi, \theta; g) = a\big].$$

We prove the proposition in two steps. Lemma 1 shows that this decision rule is obedient and achieves $TS(g)$. Lemma 2 shows that for any obedient decision rule, the expected total surplus does not exceed $TS(g)$.

Lemma 1 (Nested rule is obedient and achieves $TS(g)$)**.** *The decision rule σ^g is obedient under ownership $g \in \mathcal{N}$. Moreover, each project k succeeds with probability $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$, and hence*

$$\mathbb{E}_\psi\Big[TS(a, g; \theta)\Big] = \sum_{k \in \mathcal{K}} V_k \cdot \min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) = TS(g),$$

where $\psi(\theta, a) = p(\theta)\sigma^g(a \mid \theta)$ is the induced distribution over actions and states.

Proof. We verify obedience and compute the induced success probabilities.

Posteriors. Party i observes (θ_i, a_i) before choosing actions. The expropriation recommendations in (A.2) are constant and uninformative. Since $x_{ij} = \bar{x}_{ij}$ for all i, j in (A.3), for notational convenience we will keep track of the vector $x_i := (x_{i1}, \ldots, x_{iN})$ only.

Fix i and write $x_{i,-i} := (x_{ij})_{j\neq i}$. For each $j \neq i$, let $s_{ij}(\xi) \in \{-1, 1\}$ be a deterministic sign defined by $s_{ij}(\xi) = 1$ if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$ and $s_{ij}(\xi) = -1$ otherwise, so that $x_{ij} = s_{ij}(\xi) \cdot \theta_j$. For any vector $v \in \{-1, 1\}^{N-1}$ and any $\xi \in [0, 1]$, independence and uniformity of $(\theta_j)_{j\neq i}$ imply

$$\begin{aligned} \mathbb{P}_\psi(x_{i,-i} = v \mid \xi) &= \mathbb{P}_\psi\big(s_{ij}(\xi) \cdot \theta_j = v_j \ \forall j \neq i \,\big|\, \xi\big) \\ &= \prod_{j\neq i} \mathbb{P}_\psi\big(\theta_j = s_{ij}(\xi) \cdot v_j\big) = \prod_{j\neq i} \tfrac{1}{2} = 2^{-(N-1)}. \end{aligned} \tag{A.4}$$

This conditional probability does *not* depend on ξ. Hence $x_{i,-i}$ is independent of ξ. Moreover, since θ_i is independent of $(\xi, x_{i,-i})$, ξ is also independent of $(\theta_i, x_{i,-i})$. Equivalently, Bayes' rule yields that for any $v \in \{-1, 1\}^{N-1}$, the conditional density of $\xi \mid x_{i,-i}$ is

$$f_{\xi|x_{i,-i}}(\xi \mid v) = \frac{\mathbb{P}_\psi(x_{i,-i} = v \mid \xi)\, f_\xi(\xi)}{\mathbb{P}_\psi(x_{i,-i} = v)} = \frac{2^{-(N-1)} \cdot 1}{2^{-(N-1)}} = 1,$$

so $\xi \mid x_{i,-i}$ is uniform on $[0, 1]$. Since θ_i is independent of $(\xi, x_{i,-i})$, we also have $f_{\xi|\theta_i,x_{i,-i}}(\cdot \mid \theta_i, x_{i,-i}) := 1$.

Fix $j \neq i$. Under the nested rule, $\theta_j = x_{ij}$ if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$. Therefore, using iterated expectations and the posterior in (A.4),

$$\begin{aligned} \mathbb{P}_\psi(\theta_j = x_{ij} \mid \theta_i, x_{i,-i}) &= \mathbb{P}_\psi(\xi \leqslant \tau_{ij}^{\text{safe}}(g) \mid \theta_i, x_{i,-i}) \\ &= \int_0^{\tau_{ij}^{\text{safe}}(g)} f_{\xi|\theta_i,x_{i,-i}}(\xi \mid \theta_i, x_{i,-i})\, d\xi = \int_0^{\tau_{ij}^{\text{safe}}(g)} 1\, d\xi = \tau_{ij}^{\text{safe}}(g). \end{aligned}$$

No expropriation. The recommendation rule always prescribes no expropriation (A.2). If i is disarmed, then every effective expropriation action is 0 and there is no deviation. If i is armed, then $\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^\star = c_{ij}/(b_{ij} + c_{ij})$ by definition, so conditional on (θ_i, a_i) the maximal success probability from attempting expropriation against any target is at most $\tau_{ij}^\star$, implying that the expected private gain from expropriation is weakly non-positive. With the tie-breaking rule favoring the safe action, obeying (A.2) is optimal.

Obedient Cooperation. Now consider deviations that change i's cooperative actions $(x_i, \bar{x}_i)$. For any project k, let $J_{i,k}(g)$ denote the set of indices $j \neq i$ such that $(i, j) \in \mathcal{Q}_k(g)$, i.e., the states that i must match (via either x_{ij} if $i = g$ and $j \in \mathcal{R}_k$, or via $\bar{x}_{ij}$ if $(i, j) \in \bar{\mathcal{R}}_k$). If $J_{i,k}(g) = \varnothing$, then i's cooperative action does not affect project k.

Fix a project k where $J_{i,k}(g) \neq \varnothing$. Under the nested rule, if all parties follow their recommendations, project k succeeds if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$ for all required tasks (i, j), which is equivalent to:

$$\xi \leqslant \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g). \tag{A.5}$$

Since $\tau_{ij}^{\text{safe}}(g) > 1/2$ for all pairs, this probability is strictly greater than 1/2.

Now suppose party i deviates from its recommendation for a subset of targets $\tilde{J}_{i,k}(g) \subseteq J_{i,k}(g)$. Party i successfully matches the states if and only if the recommendation was correct for the non-deviated targets ($j \notin \tilde{J}_{i,k}(g)$) and incorrect for the deviated targets ($j \in \tilde{J}_{i,k}(g)$). In terms of the randomization variable ξ, this requires:

$$\xi \in \Big(\max_{j\in\tilde{J}_{i,k}(g)} \tau_{ij}^{\text{safe}}(g), \min_{j\in J_{i,k}(g)\setminus\tilde{J}_{i,k}(g)} \tau_{ij}^{\text{safe}}(g) \Big], \quad \text{with the convention } \min\varnothing := 1.$$

Since $\tau_{ij}^{\text{safe}}(g) > 1/2$, the lower bound of this interval is strictly greater than 1/2. Therefore, the length of the interval—and thus the success probability—is strictly less than 1/2. Consequently, any deviation yields a strictly lower probability of success than obedience.

Total surplus. Multiplying the probability of the event in (A.5) by V_k and summing over all projects yields the stated expression for $TS(g)$. □

Obedient joint distributions. The next lemma establishes an upper bound on total surplus that applies to any obedient decision rule. To state it in sufficient generality, we first define obedience on joint distributions over states and actions. A joint distribution $\psi \in \Delta(\mathcal{A} \times \Theta)$ is *obedient under owner g* if, for every $i \in \mathcal{N}$, $\theta_i \in \Theta_i$, $a_i \in \mathcal{A}_i$, and $a_i' \in \mathcal{A}_i$,

$$\sum_{\theta_{-i}\in\Theta_{-i}} \sum_{a_{-i}\in\mathcal{A}_{-i}} \psi(\theta_i, \theta_{-i}, a_i, a_{-i}) \Big[u_i((a_i, a_{-i}), g; \theta) - u_i((a_i', a_{-i}), g; \theta) \Big] \geqslant 0. \tag{A.6}$$

The obedience condition for a decision rule σ in (3.1) is a special case: any obedient σ induces an obedient joint distribution $\psi(a, \theta) = p(\theta)\sigma(a \mid \theta)$.

Lemma 2 (Upper bound)**.** *Fix ownership $g \in \mathcal{N}$. For any obedient joint distribution $\psi \in \Delta(\mathcal{A} \times \Theta)$,*

$$\mathbb{E}_\psi[TS(a, g; \theta)] \leqslant \sum_{k\in\mathcal{K}} V_k \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g).$$

Proof. For each task $(i, j) \in \mathcal{Q}_k(g)$ define the event that this task is satisfied:

$$C_{k,(i,j)} := \begin{cases} \{(\theta, a) : x_{ij} = \theta_j\} & \text{if } (i,j) = (g,j) \in \{(g,j) : j \in \mathcal{R}_k\} \text{ and } (i,j) \notin \bar{\mathcal{R}}_k, \\ \{(\theta, a) : \bar{x}_{ij} = \theta_j\} & \text{if } (i,j) \in \bar{\mathcal{R}}_k \text{ and } (i,j) \notin \{(g,j) : j \in \mathcal{R}_k\}, \\ \{(\theta, a) : x_{ij} = \bar{x}_{ij} = \theta_j\} & \text{if } (i,j) = (g,j) \in \{(g,j) : j \in \mathcal{R}_k\} \cap \bar{\mathcal{R}}_k. \end{cases} \tag{A.7}$$

Let E_k denote the event that project k succeeds:

$$E_k := \bigcap_{(i,j)\in\mathcal{Q}_k(g)} C_{k,(i,j)}.$$

We write total surplus as

$$TS(a,g;\theta) = TS^{\text{coop}}(a,g;\theta) + TS^{\text{exp}}(a,g;\theta),$$

where $TS^{\text{coop}}(a,g;\theta) = \sum_{k\in\mathcal{K}} V_k\,\mathbf{1}\{E_k\}$, and TS^{exp} is the sum of all expropriation terms.

Upper Bound on Task Probabilities. Let

$$r_{ij} := \mathbb{P}_\psi\left(\zeta_{ij} \neq 0\right)$$

be the probability that party i attempts an expropriation of party j's information, $j \neq i$. If $i \notin \text{Arm}(g)$, then party i cannot engage in expropriation and $r_{ij} = 0$.

Decompose $\mathbb{P}_\psi(C_{k,(i,j)})$ according to whether i attempts to expropriate j:

$$\mathbb{P}_\psi(C_{k,(i,j)}) = \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} = 0) + \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} \neq 0).$$

On the event $\{\zeta_{ij} = 0\}$, obedience implies that attempting expropriation is not strictly profitable for party i. Therefore, conditional on (θ_i, a_i), the probability that any sign choice satisfies the matching task is at most $\tau_{ij}^{\text{safe}}(g)$, so

$$\mathbb{P}_\psi\left(C_{k,(i,j)} \mid \zeta_{ij} = 0\right) \leqslant \tau_{ij}^{\text{safe}}(g) \;\Rightarrow\; \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} = 0) \leqslant \tau_{ij}^{\text{safe}}(g)\mathbb{P}_\psi\left(\zeta_{ij} = 0\right) = \tau_{ij}^{\text{safe}}(g)(1 - r_{ij}).$$

In addition,

$$\mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} \neq 0) \leqslant \mathbb{P}_\psi\left(\zeta_{ij} \neq 0\right) = r_{ij}.$$

Combining these two inequalities, we get

$$\mathbb{P}_\psi(C_{k,(i,j)}) \leqslant \tau_{ij}^{\text{safe}}(g)(1 - r_{ij}) + r_{ij}.$$

Upper bound on cooperative surplus. Fix project $k \in \mathcal{K}$. If $\mathcal{Q}_k(g)$ contains only trivial tasks of the form (i,i), then $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) = 1$, so $\mathbb{P}_\psi\left(E_k\right) \leqslant \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$.

Fix any project k with non-trivial tasks: $(i,j) \in \mathcal{Q}_k(g)$ with $i \neq j$. Pick one bottleneck task:

$$(i_k, j_k) \in \arg\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \quad \text{with } j_k \neq i_k.$$

Since $E_k \subseteq C_{k,i_k,j_k}$, we have

$$\mathbb{P}_\psi(E_k) \leqslant \tau^{\text{safe}}_{i_k j_k}(g) + \big(1-\tau^{\text{safe}}_{i_k j_k}(g)\big) r_{i_k j_k},$$

Multiplying by V_k and summing over projects yields

$$\mathbb{E}_\psi[TS^{\text{coop}}] \leqslant TS(g) + \sum_{i\in \text{Arm}(g)} \sum_{j\neq i} \big(1-\tau^{\text{safe}}_{ij}(g)\big)\, V^{\min}_{ij}\, r_{ij}, \tag{A.8}$$

where

$$V^{\min}_{ij} := \sum_{k:\ (i_k,j_k)=(i,j)} V_k.$$

Upper bound on expropriation surplus. Fix a pair (i,j) with $i \in \text{Arm}(g)$ and $j \neq i$. On the event $\{\zeta_{ij} \neq 0\}$, obedience implies that attempting expropriation is (weakly) optimal, hence the conditional success probability is at least the private threshold:

$$\mathbb{P}_\psi(\zeta_{ij} = \theta_j \mid \zeta_{ij} \neq 0) \ \geqslant\ \tau^\star_{ij}.$$

Therefore the expected welfare contribution from this ordered pair satisfies

$$\mathbb{E}_\psi\Big[(b_{ij}-\lambda_{ij})\,\mathbf{1}\{\zeta_{ij}=\theta_j\} - c_{ij}\,\mathbf{1}\{\zeta_{ij}=-\theta_j\}\Big] \leqslant \mathbb{E}_\psi\Big[(b_{ij}-\lambda_{ij})\,\mathbf{1}\{\zeta_{ij}=\theta_j\}\Big] \leqslant (b_{ij}-\lambda_{ij})\,\tau^\star_{ij}\, r_{ij},$$

where the first inequality follows from $-c_{ij} < 0$ and the second inequality follows from the fact that the probability of successful expropriation should be at least $\tau^\star_{ij}$. Summing over (i,j) gives

$$\mathbb{E}_\psi[TS^{\text{exp}}] \leqslant \sum_{i\in \text{Arm}(g)} \sum_{j\neq i} \tau^\star_{ij}\,(b_{ij}-\lambda_{ij})\, r_{ij}. \tag{A.9}$$

Upper bound on total surplus. Combining (A.8) and (A.9) gives

$$\begin{aligned}\mathbb{E}_\psi[TS(a,g;\theta)] &\leqslant TS(g) + \sum_{i\in \text{Arm}(g)} \sum_{j\neq i} r_{ij}\Big[(1-\tau^\star_{ij})V^{\min}_{ij} + \tau^\star_{ij}(b_{ij}-\lambda_{ij})\Big] \\ &= TS(g) - \sum_{i\in \text{Arm}(g)} \sum_{j\neq i} r_{ij}\Big[\underbrace{\tau^\star_{ij}(\lambda_{ij}-b_{ij}) - \big(1-\tau^\star_{ij}\big)V^{\min}_{ij}}_{>0}\Big].\end{aligned}$$

By Assumption 1, $\lambda_{ij} - b_{ij} > V_{\text{total}} \geqslant V^{\min}_{ij}$. Also $b_{ij} < c_{ij}$ implies $\tau^\star_{ij} = \frac{c_{ij}}{b_{ij}+c_{ij}} > \frac{1}{2}$, hence $\tau^\star_{ij} > (1-\tau^\star_{ij})$. Therefore, for every $i \in \text{Arm}(g)$,

$$\tau^\star_{ij}(\lambda_{ij}-b_{ij}) \ >\ \tau^\star_{ij}V^{\min}_{ij} \ \geqslant\ (1-\tau^\star_{ij})V^{\min}_{ij},$$

so the bracketed term is strictly positive. Therefore, the maximum attainable expected total surplus is $TS(g)$ and can be attained only when there is no expropriation, $r_{ij} = 0$ for all $i \in \mathcal{N}$ and $j \neq i$. $\square$

Conclusion. Lemma 1 shows $TS(g)$ is attainable by the nested no-expropriation rule. Lemma 2 shows no obedient rule can exceed it under Assumptions 1–2. This completes the proof of Proposition 1.

A.2 Implementation Through a Mediated Contract

This appendix proves the results stated in Section 5.1. We first analyze the communication game and prove Proposition 2. We then analyze the contracting game and show how the outcome from the baseline solution concept can be supported as a Perfect Bayesian Equilibrium (PBE).

A.2.1 Communication Game and Proof of Proposition 2

Incentive Constraints. Fix a direct mechanism (φ, t). When all parties report truthfully and follow their recommendations, party i's interim expected utility is

$$U_i(\varphi, t \mid \theta_i) \;:=\; \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a \in \mathcal{A}} \sum_{g \in \mathcal{N}} \varphi(a, g \mid \theta_i, \theta_{-i}) \Big[u_i(a, g; \theta_i, \theta_{-i}) + t_i\big((a, g), (\theta_i, \theta_{-i})\big) \Big].$$

Parties are not compelled to follow recommendations; a (pure) deviation for party i is a function $\delta_i : D_i \to \mathcal{A}_i$ mapping the available information at the action stage into an action plan, where $D_i := \mathcal{N} \times \mathcal{T} \times \mathcal{A}_i \times \Theta_i$, and party i observes $d_i := (g, t', a_i, \theta_i) \in D_i$. For any alternative report $\theta_i' \in \Theta_i$ and any deviation function $\delta_i : D_i \to \mathcal{A}_i$, define the *double-deviation* payoff

$$\begin{aligned} U_i^\star(\varphi, t, \delta_i, \theta_i' \mid \theta_i) \;:=\; \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a \in \mathcal{A}} \sum_{g \in \mathcal{N}} \varphi(a, g \mid \theta_i', \theta_{-i}) \Big[& u_i\big((\delta_i(g, t', a_i, \theta_i), a_{-i}), g; \theta_i, \theta_{-i}\big) \\ & + t_i\big((a, g), (\theta_i', \theta_{-i})\big) \Big], \end{aligned}$$

where $t' := t((a, g), (\theta_i', \theta_{-i}))$ is the transfer vector implemented after the report (θ_i', θ_{-i}).

The contract (φ, t) is *incentive compatible* if, for every $i \in \mathcal{N}$,

$$U_i(\varphi, t \mid \theta_i) \;\geqslant\; U_i^\star(\varphi, t, \delta_i, \theta_i' \mid \theta_i) \qquad \text{for all } \theta_i, \theta_i' \in \Theta_i \text{ and all } \delta_i : D_i \to \mathcal{A}_i. \tag{A.10}$$

This condition subsumes truth-telling (take $\delta_i(d_i) = a_i$) and obedience (take $\theta_i' = \theta_i$) as special cases.

Communication-Game Problem. The maximal expected total surplus attainable in the communication game is

$$\begin{aligned}
\max_{\varphi,t} \quad & \sum_{\theta\in\Theta} p(\theta) \sum_{g\in\mathcal{N}} \sum_{a\in\mathcal{A}} \varphi(a,g \mid \theta)\, TS(a,g;\theta) \\
\text{subject to} \quad & \text{(A.10)}, \\
& \varphi(a,g \mid \theta) \geqslant 0, \qquad \sum_{a,g} \varphi(a,g \mid \theta) = 1 \qquad \text{for all } \theta \in \Theta, \\
& \sum_{i\in\mathcal{N}} t_i(a,g,\theta) = 0 \qquad \text{for all } (a,g,\theta) \in \mathcal{A} \times \mathcal{N} \times \Theta.
\end{aligned} \tag{A.11}$$

Relaxed Problem: Obedience Only. Consider the relaxation of (A.11) obtained by dropping the truth-telling and the double-deviation constraints. In this relaxation, we evaluate incentives under the *truthful* report profile and impose only obedience of the recommended non-contractible actions.

Because transfers are implemented before actions are chosen and do not depend on realized non-contractible actions, they cancel from the obedience constraint when comparing a recommended action to a deviation. The obedience constraints are: for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every $g \in \mathcal{N}$, every recommended action plan $a_i \in \mathcal{A}_i$, and every deviation $a_i' \in \mathcal{A}_i$,

$$\sum_{\theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i}} \varphi\left(a_i, a_{-i}, g \mid \theta_i, \theta_{-i}\right) \left[u_i\left(a_i, a_{-i}, g;\theta\right) - u_i\left(a_i', a_{-i}, g;\theta\right)\right] \geqslant 0. \tag{A.12}$$

Thus, the relaxed problem is

$$\begin{aligned}
\max_{\varphi} \quad & \sum_{\theta\in\Theta} p(\theta) \sum_{g\in\mathcal{N}} \sum_{a\in\mathcal{A}} \varphi(a,g \mid \theta)\, TS(a,g;\theta) \\
\text{subject to} \quad & \text{(A.12)}, \qquad \varphi(a,g \mid \theta) \geqslant 0, \quad \sum_{a,g} \varphi(a,g \mid \theta) = 1 \ \text{ for all } \theta \in \Theta.
\end{aligned} \tag{A.13}$$

For any fixed ownership assignment $g \in \mathcal{N}$, if we restrict attention to allocation rules that put probability one on g, then (A.13) coincides with the benchmark mediator problem (2.1) defining $TS(g)$.

Lemma 3 (Deterministic ownership)**.** *In the communication-game problem* (A.13)*, allowing randomization of ownership g does not increase the maximal attainable total surplus. There exists an optimal solution with deterministic ownership.*

Proof. Let φ be any feasible allocation rule for (A.13), and let $G \in \mathcal{N}$ denote the (possibly random) owner induced by φ (which is publicly observed before actions are chosen). Let $\psi(a,g,\theta) := p(\theta)\varphi(a,g \mid \theta)$ denote the joint distribution of (a,g,θ) induced by φ.

Fix any $g \in \mathcal{N}$ with $\mathbb{P}_\psi(G = g) > 0$, and define the conditional joint distribution of actions and states given $G = g$ by

$$\psi^g(a, \theta) := \mathbb{P}_\psi(a, \theta \mid G = g) = \frac{\psi(a, g, \theta)}{\mathbb{P}_\psi(G = g)}.$$

Because the obedience constraints (A.12) are imposed separately for each ownership realization g, and because parties observe $G = g$ before choosing actions, the conditional distribution ψ^g is obedient under owner g. Therefore, by Lemma 2,

$$\mathbb{E}_{\psi^g}\big[TS(a, g; \theta)\big] \leqslant TS(g).$$

Taking expectations over G yields

$$\mathbb{E}_\psi[TS(a, G; \theta)] = \sum_{g \in \mathcal{N}} \mathbb{P}_\psi(G = g)\, \mathbb{E}_{\psi^g}[TS(a, g; \theta)] \leqslant \sum_{g \in \mathcal{N}} \mathbb{P}_\psi(G = g)\, TS(g) \leqslant \max_{g \in \mathcal{N}} TS(g).$$

Finally, choosing deterministic ownership $G \equiv g^* \in \arg\max_{g \in \mathcal{N}} TS(g)$ and an optimal obedient decision rule that attains $TS(g^*)$ achieves the upper bound. Hence an optimal solution to (A.13) exists with deterministic ownership. $\square$

The solution to the relaxed (obedience-only) problem in (2.1) with deterministic ownership provides an upper bound for the communication-game problem (A.11). Let σ^* be an optimal decision rule for the problem in (2.1), and let g^* denote an optimal owner, $g^* \in \arg\max_g TS(g)$.

Lemma 4 (Constant-transfer implementation)**.** *Fix $g^* \in \arg\max_{g \in \mathcal{N}} TS(g)$ and let σ^* be the nested optimal decision rule constructed in Proposition 1 under owner g^*. Then for any constant budget-balanced transfer vector $\bar{t} \in \mathcal{T}$, the direct mediated mechanism $(\varphi^*, \bar{t})$ defined by*

$$\varphi^*(a, g \mid \theta) = \begin{cases} \sigma^*(a \mid \theta), & \text{if } g = g^*, \\ 0, & \text{if } g \neq g^*, \end{cases} \tag{A.14}$$

is incentive compatible.

Proof. Because transfers are constant, they cancel from all report and deviation comparisons. Fix a party i, a true type θ_i, an alternative report $\hat{\theta}_i \in \Theta_i$, and an arbitrary action-stage deviation $\delta_i : D_i \to \mathcal{A}_i$.

Step 1: Upper Bound for a Single Task. Fix any target $j \neq i$. Under truthful reporting by party j, we have $\hat{\theta}_j = \theta_j$, and party i receives a recommendation

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j, & \xi \leqslant \tau_{ij}^{\mathrm{safe}}(g^*), \\ -\theta_j, & \xi > \tau_{ij}^{\mathrm{safe}}(g^*). \end{cases} \tag{A.15}$$

In Lemma 1, we proved that the posterior belief that $\theta_j = x_{ij}$ under (A.15) is equal to $\tau_{ij}^{\mathrm{safe}}(g^*)$. Since $\tau_{ij}^{\mathrm{safe}}(g^*) \geqslant \frac{1}{2}$, x_{ij} is a posterior mode for θ_j given d_i. Thus, among all choices of i's productive action for target j, the maximal achievable success probability in task (i,j) equals $\tau_{ij}^{\mathrm{safe}}(g^*)$. Therefore, for any deviation δ_i and any report $\hat{\theta}_i$, the probability that task (i,j) in project k succeeds (event $C_{k,(ij)}$ defined in (A.7)) has the following upper bound:

$$\mathbb{P}(C_{k,(ij)} \mid \hat{\theta}_i, \delta_i, \theta_i) \ \leqslant\ \tau_{ij}^{\mathrm{safe}}(g^*) \qquad \text{for all } j \neq i. \tag{A.16}$$

A similar bound applies to tasks performed by *other* parties who obey. Fix $\ell \neq i$ and $j \neq \ell$. Since ℓ is held to obey equilibrium recommendations, its match success satisfies

$$\mathbb{P}(C_{k,(\ell j)} \mid \hat{\theta}_i, \delta_i, \theta_i) = \begin{cases} \tau_{\ell j}^{\mathrm{safe}}(g^*), & j \neq i \text{ (since } \hat{\theta}_j = \theta_j), \\ \tau_{\ell i}^{\mathrm{safe}}(g^*), & j = i,\ \hat{\theta}_i = \theta_i, \\ 1 - \tau_{\ell i}^{\mathrm{safe}}(g^*), & j = i,\ \hat{\theta}_i = -\theta_i. \end{cases}$$

Furthermore, since $\tau_{\ell i}^{\mathrm{safe}}(g^*) \geqslant \frac{1}{2}$, in all cases we have

$$\mathbb{P}(C_{k,(\ell j)} \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \tau_{\ell j}^{\mathrm{safe}}(g^*). \tag{A.17}$$

Step 2: Upper Bound on Project Success. For each project k, success requires all required tasks to succeed: $E_k = \bigcap_{(\ell,j)\in\mathcal{Q}_k(g^*)} C_{k,(\ell j)}$. Hence for any $\hat{\theta}_i, \delta_i$,

$$\mathbb{P}(E_k \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \min_{(\ell,j)\in\mathcal{Q}_k(g^*)} \mathbb{P}(C_{k,(\ell j)} \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \min_{(\ell,j)\in\mathcal{Q}_k(g^*)} \tau_{\ell j}^{\mathrm{safe}}(g^*).$$

Under truthful reporting and obedience, each required task (ℓ, j) succeeds if and only if $\xi \leqslant \tau_{\ell j}^{\mathrm{safe}}(g^*)$, so (because the same ξ is shared across tasks)

$$\mathbb{P}(E_k \mid \text{truth, obey}, \theta_i) = \min_{(\ell,j)\in\mathcal{Q}_k(g^*)} \tau_{\ell j}^{\mathrm{safe}}(g^*).$$

That is, truthful reporting and obedience attain the upper bound for every project.

Since $u_i^c = \sum_{k\in\mathcal{K}} \alpha_{ik} V_k\, \mathbf{1}\{E_k\}$ with $\alpha_{ik} V_k \geqslant 0$, maximizing each $\mathbb{P}(E_k)$ implies that truthful reporting and obedience maximize i's expected cooperative payoff.

No Expropriation. Under σ^* the posterior accuracy about any target j is capped at $\tau_{ij}^{\text{safe}}(g^*)$, so by the definition of the safety threshold (see Lemma 1), any expropriation attempt yields a weakly lower expected payoff than $\zeta_{ij} = 0$. The expropriation payoff is unaffected by i's report because i's information about θ_{-i} is independent of $\hat{\theta}_i$.

Conclusion. Combining the cooperative and expropriation parts yields the double-deviation IC constraints (A.10). $\square$

Proposition 2. Lemma 3 shows that $TS(g^*)$ provides an upper bound on the maximal surplus in the communication game. Lemma 4 shows that this surplus is attainable. Together, these results complete the proof of Proposition 2.

A.2.2 Contracting Game

We now embed the communication game into a broader bargaining environment. Fix an initial owner $g_0 \in \mathcal{N}$. A *contract* is a direct mediated mechanism (φ, t). The contracting game adds two stages before the communication game:

1. Party 1 makes a take-it-or-leave-it offer of a contract (φ, t).
2. Each party $i \in \{2, \dots, N\}$ either accepts or rejects.

If all parties accept, the communication game induced by (φ, t) is played. If at least one party rejects, the contract is void, the owner remains g_0, transfers are zero, no information is shared, and the disagreement outcome is played.

Let (a^0, g_0) denote the disagreement outcome, where $x_{ii} = \bar{x}_{ii} = \theta_i$ for all i, $x_{ij} = \bar{x}_{ij} = 1$ for all $j \neq i$, and $z_{ij} = \bar{z}_{ij} = 0$ for all $j \neq i$. Let

$$\underline{U}_i := \sum_{\theta \in \Theta} p(\theta)\, u_i(a^0, g_0; \theta)$$

denote party i's ex-ante disagreement payoff. Let

$$\bar{U}_i^* := \sum_{\theta \in \Theta} p(\theta) \sum_{a \in \mathcal{A}} \sigma^*(a \mid \theta)\, u_i(a, g^*; \theta)$$

denote party i's expected non-transfer payoff under the optimal benchmark rule. Define the constant transfer vector $t^* \in \mathcal{T}$ by

$$t_i^* = \underline{U}_i - \bar{U}_i^* \quad (i \neq 1), \qquad t_1^* = -\sum_{i \neq 1} t_i^*.$$

Let (φ^*, t^*) denote the associated mediated contract, where φ^* is defined in (A.14).

Proposition 5 (PBE implementation of the optimal contract)**.** *There exists a Perfect Bayesian Equilibrium of the full bargaining game in which:*

1. *Party 1 offers the contract* (φ^*, t^*)*.*

2. *Every party* $i \neq 1$ *accepts.*

3. *Conditional on acceptance of* (φ^*, t^*)*, every party reports truthfully and obeys its recommendation on the equilibrium path.*

The induced on-path joint distribution over ownership, states, and actions is

$$p(\theta)\,\mathbf{1}\{g = g^*\}\sigma^*(a \mid \theta),$$

and total surplus equals $\max_{g \in \mathcal{N}} TS(g)$*.*

Proof. For each (φ, t) among feasible direct mediated contracts, the continuation (communication) game induced by (φ, t) is a finite extensive-form game with perfect recall. Hence it has a sequential equilibrium (Kreps and Wilson, 1982), and therefore at least one PBE. For each feasible direct contract (φ, t), fix one continuation PBE of the induced communication game, and let $W_i(\varphi, t)$ denote party i's ex-ante payoff in that continuation equilibrium. For the optimal contract (φ^*, t^*), choose the truthful-reporting and obedient continuation Bayesian Nash equilibrium from Lemma 4, and we complete it to a continuation PBE using the off-path beliefs specified below.

Define the stage-1 offer strategy of party 1 by offering (φ^*, t^*). For each $i \neq 1$, define the stage-2 acceptance rule by

$$\chi_i(\varphi, t) = \begin{cases} \text{accept}, & \text{if } W_i(\varphi, t) \geqslant \underline{U}_i, \\ \text{reject}, & \text{if } W_i(\varphi, t) < \underline{U}_i. \end{cases}$$

Because t^* was chosen so that

$$W_i(\varphi^*, t^*) = \underline{U}_i \qquad \text{for every } i \neq 1,$$

every party $i \neq 1$ accepts (φ^*, t^*).

Conditional on acceptance of (φ^*, t^*), let every party report truthfully and obey its recommendation on path. By Lemma 4, this continuation behavior is a Bayesian Nash equilibrium of the communication game induced by (φ^*, t^*). Since the prior has full support, every report profile to the mediator occurs with positive probability. Consider any action-stage information set of party i, $d_i = (g, \tilde{t}, a_i, \theta_i) \in D_i$, that has zero probability under the equilibrium path. We distinguish two cases.

First, d_i may be *report-consistent*: there exists a report profile $\hat{\theta} \in \Theta$ and an action profile $a = (a_i, a_{-i})$ such that $\varphi^*(a, g \mid \hat{\theta}) > 0$ and $\tilde{t} = t^*((a, g), \hat{\theta})$. At such histories, beliefs are determined by Bayes' rule conditional on d_i and on party i's information about its own report. Specifically, if i reported truthfully, beliefs are conditioned on truthful reporting by all parties; if i misreported, beliefs are conditioned on i's known report $\hat{\theta}_i$ and truthful reporting by the others. The continuation action of party i is then chosen as a best response to those beliefs.

Second, d_i may be *report-inconsistent*: there is no report profile $\hat{\theta} \in \Theta$ and no action profile $a = (a_i, a_{-i})$ such that $\varphi^*(a, g \mid \hat{\theta}) > 0$ and $\tilde{t} = t^*((a, g), \hat{\theta})$. These are impossible histories under the fixed contract (φ^*, t^*). At such histories, beliefs are unrestricted by Bayes' rule, and we assign the belief μ_i described below.

For each report-inconsistent information set d_i and each $j \neq i$, define a productive sign $\tilde{s}_{ij}(d_i) \in \{-1, 1\}$ by

$$\tilde{s}_{ij}(d_i) := \begin{cases} x_{ij}, & \text{if } x_{ij} = \bar{x}_{ij} \in \{-1, 1\}, \\ 1, & \text{otherwise.} \end{cases}$$

At d_i, prescribe the continuation actions $z_{ij} = \bar{z}_{ij} = 0$ for all $j \neq i$, $x_{ii} = \bar{x}_{ii} = \theta_i$, and $x_{ij} = \bar{x}_{ij} = \tilde{s}_{ij}(d_i)$ for $j \neq i$. Party i's off-path belief $\mu_i(\cdot \mid d_i)$ over θ_{-i} is generated by a latent common random variable $\xi \sim \text{Unif}[0, 1]$ according to

$$\theta_j = \begin{cases} \tilde{s}_{ij}(d_i), & \text{if } \xi \leqslant \tau_{ij}^{\text{safe}}(g), \\ -\tilde{s}_{ij}(d_i), & \text{if } \xi > \tau_{ij}^{\text{safe}}(g), \end{cases} \qquad \text{for every } j \neq i.$$

Moreover, party i believes that the same latent ξ governs all other parties' productive recommendations under owner g, and that all other parties follow their continuation strategies. Under these beliefs, each sign $\tilde{s}_{ij}(d_i)$ is a posterior mode for θ_j, and the posterior precision is $\tau_{ij}^{\text{safe}}(g)$. Hence no expropriation is weakly optimal, and the prescribed productive action plan is sequentially rational.

Finally, by Proposition 2, no direct contract can induce a continuation surplus strictly above $\max_{g \in \mathcal{N}} TS(g)$, whereas (φ^*, t^*) attains that value. If all parties accept a contract (φ, t), then

$$W_1(\varphi, t) = \sum_{i=1}^{N} W_i(\varphi, t) - \sum_{i \neq 1} W_i(\varphi, t) \leqslant \max_{g \in \mathcal{N}} TS(g) - \sum_{i \neq 1} \underline{U}_i = W_1(\varphi^*, t^*),$$

where the inequality uses Proposition 2 and the fact that acceptance implies $W_i(\varphi, t) \geqslant \underline{U}_i$ for all $i \neq 1$. If some party rejects (φ, t), party 1 receives the disagreement payoff, which is weakly below $W_1(\varphi^*, t^*)$. Hence party 1 weakly prefers offering (φ^*, t^*). Therefore the offer strategy, acceptance rules, continuation strategies, and belief system constructed above define a PBE of the full bargaining game. $\square$

A.3 Contractible Actions and Cooperative Payoffs

In the baseline model, all components of the action plan $a_i = (x_i, \bar{x}_i, z_i, \bar{z}_i) \in \mathcal{A}_i$ are *non-contractible*: the mediator can only send private recommendations, and obedience must be verified against deviations in *all* components. In addition, project shares $\{\alpha_{ik}\}_{i,k}$ are fixed and exogenously given.

This section studies three counterfactual environments. First, we allow expropriation actions to be contractible; this makes it possible to implement the first best. Second, we allow partial contractibility of expropriation through a verifiable signal and a penalty transfer. Third, we allow cooperative actions and project shares to be contractible. We show that making cooperative actions and shares contractible does not increase the maximum attainable surplus relative to the baseline.

For notational convenience, decompose each party's action plan into a *cooperation block* and an *expropriation block*:

$$a_i^c := (x_i, \bar{x}_i) \in \mathcal{A}_i^c = \{-1, 1\}^{2N} \qquad a_i^e := (z_i, \bar{z}_i) \in \mathcal{A}_i^e = \{-1, 0, 1\}^{2(N-1)},$$

so that $\mathcal{A}_i = \mathcal{A}_i^c \times \mathcal{A}_i^e$ and $a_i = (a_i^c, a_i^e)$. The mediator observes θ and draws an action profile $a = (a_i)_{i\in\mathcal{N}} \in \mathcal{A}$ from a direct decision rule $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$.

(i) Contractible expropriation actions. Suppose the mediator can *enforce* expropriation actions $(z, \bar{z})$, while cooperative actions $(x, \bar{x})$ remain non-contractible. Then, after receiving a recommendation $a_i = (a_i^c, a_i^e)$, party i can deviate only in the cooperation block a_i^c; the expropriation block a_i^e is fixed.

Accordingly, σ is *obedient under g with contractible expropriation* if, for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every recommended $a_i = (a_i^c, a_i^e) \in \mathcal{A}_i$, and every alternative $\tilde{a}_i^c \in \mathcal{A}_i^c$,

$$\sum_{\theta_{-i}\in\Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i}\in\mathcal{A}_{-i}} \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big(((a_i^c, a_i^e), a_{-i}), g; \theta\big) - u_i\big(((\tilde{a}_i^c, a_i^e), a_{-i}), g; \theta\big)\Big] \geqslant 0. \tag{A.18}$$

Define the corresponding omniscient benchmark value:

$$TS^e(g) := \max_{\sigma} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.19}$$
$$\text{subject to (A.18)}, \quad \sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta$$

(ii) Partial contractibility of expropriation. Suppose that expropriation actions themselves cannot be contracted on, but that after actions are chosen there is a verifiable signal $s^e_{ij} \in \{0,1\}$ indicating an expropriation attempt by party i against party j. If $\zeta_{ij} \neq 0$, then $s^e_{ij} = 1$ with probability $\rho^e_{ij} \in [0,1]$; if $\zeta_{ij} = 0$, then $s^e_{ij} = 0$ surely. In addition to choosing a decision rule σ, the mediator specifies penalty transfers $\{t^e_{ij}\}_{i \neq j}$ with $t^e_{ij} \in [0, \bar{t}_{ij}]$. If the signal realizes $s^e_{ij} = 1$, then i transfers t^e_{ij} to j. Cooperative and expropriation actions remain non-contractible.

Accordingly, σ is *obedient under g with partially contractible expropriation* if, for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every recommended $a_i = (a^c_i, a^e_i) \in \mathcal{A}_i$, and every alternative $\tilde{a}_i = (\tilde{a}^c_i, \tilde{a}^e_i) \in \mathcal{A}_i$,

$$\begin{aligned}\sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big((a_i, a_{-i}), g; \theta\big) - \sum_{j \neq i} \rho^e_{ij} t^e_{ij} \mathbf{1}\{\zeta_{ij}(a_i; g) \neq 0\} \\ -u_i\big((\tilde{a}_i, a_{-i}), g; \theta\big) + \sum_{j \neq i} \rho^e_{ij} t^e_{ij} \mathbf{1}\{\zeta_{ij}(\tilde{a}_i; g) \neq 0\} \Big] \geqslant 0.\end{aligned} \tag{A.20}$$

The corresponding benchmark value is

$$TS^{\text{partial}}(g) := \max_{\sigma, \{t^e_{ij}\}} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.21}$$
$$\text{subject to (A.20)}, \quad 0 \leqslant t^e_{ij} \leqslant \bar{t}_{ij} \text{ for all } i \neq j,$$
$$\sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta.$$

(iii) Contractible actions and cooperative payoffs. Suppose instead that the mediator can *enforce* cooperative actions $(x, \bar{x})$, while expropriation actions $(z, \bar{z})$ remain non-contractible. Then party i can deviate only in the expropriation block a^e_i; the cooperation block a^c_i is fixed.

Accordingly, σ is *obedient under g with contractible cooperation* if, for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every recommended $a_i = (a^c_i, a^e_i) \in \mathcal{A}_i$, and every alternative $\tilde{a}^e_i \in \mathcal{A}^e_i$,

$$\sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big(((a^c_i, a^e_i), a_{-i}), g; \theta\big) - u_i\big(((a^c_i, \tilde{a}^e_i), a_{-i}), g; \theta\big) \Big] \geqslant 0. \tag{A.22}$$

Suppose the mediator can also choose and enforce the division of cooperative payoffs $\{\alpha_{ik}\}_{i,k}$.

Thus, the corresponding omniscient mediator's benchmark value is

$$TS^c(g) := \max_{\sigma,\{\alpha_{ik}\}} \sum_\theta p(\theta) \sum_a \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.23}$$
$$\text{subject to (A.22)}, \quad \alpha_{ik} \geqslant 0, \quad \sum_{i\in\mathcal{N}} \alpha_{ik} = 1 \text{ for all } k \in \mathcal{K},$$
$$\sigma(a \mid \theta) \geqslant 0, \quad \sum_a \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta.$$

Proposition 6 (Contractibility and the attainable surplus)**.** *Fix any ownership assignment $g \in \mathcal{N}$. If expropriation actions $(z, \bar{z})$ are contractible, then*

$$TS^e(g) = V_{\text{total}} \qquad \textit{for every } g \in \mathcal{N}.$$

If expropriation is partially contractible with detection probabilities (ρ^e_{ij}) and enforceable transfer bounds $(\bar{t}_{ij})$, then

$$TS^{\text{partial}}(g) = \sum_{k\in\mathcal{K}} V_k \min_{(i,j)\in\mathcal{Q}_k(g)} \tilde{\tau}^{\text{safe}}_{ij}(g),$$

where the modified safety threshold is defined by

$$\tilde{\tau}^{\text{safe}}_{ij}(g) := \begin{cases} \min\Big\{1, \dfrac{c_{ij} + \rho^e_{ij}\bar{t}_{ij}}{b_{ij} + c_{ij}}\Big\} & \textit{if } i \in \text{Arm}(g), \\ 1 & \textit{if } i \notin \text{Arm}(g). \end{cases}$$

If cooperative actions $(x, \bar{x})$ and project shares $\{\alpha_{ik}\}_{i,k}$ are contractible, then

$$TS^c(g) = TS(g) \qquad \textit{for every } g \in \mathcal{N},$$

where $TS(g)$ is the value characterized in Proposition 1.

Proof. **(i) Contractible expropriation actions:** $TS^e(g) = V_{\text{total}}$. Define a (deterministic) decision rule σ^{FB} as follows. For every $\theta \in \Theta$:

1. Set *all* expropriation actions to zero: $z_{ij} = 0$ and $\bar{z}_{ij} = 0$ for all i and all $j \neq i$;
2. Recommend $x_{ij} = \bar{x}_{ij} = \theta_j$ for all $i \in \mathcal{N}$ and $j \in \mathcal{N}$.

Under σ^{FB}, there is no expropriation, and every project succeeds surely. Thus the total surplus is V_{total}. Since $(z, \bar{z})$ are enforced, party i can deviate only in $a^c_i = (x_i, \bar{x}_i)$. Any deviation in a^c_i can only leave project success unchanged or violate some matching requirement that uses i's action; since cooperative payoffs have nonnegative shares, such a deviation cannot increase i's payoff. Thus, σ^{FB} is obedient.

(ii) Partial contractibility of expropriation: $TS^{\text{partial}}(g) = \sum_k V_k \min_{(i,j)\in\mathcal{Q}_k(g)} \tilde{\tau}^{\text{safe}}_{ij}(g)$.

Fix any penalty vector t^e with $0 \leqslant t^e_{ij} \leqslant \bar{t}_{ij}$. If party i attempts to expropriate party j's information and the attempt succeeds with posterior probability τ, its expected payoff from the attempt, net of the expected penalty, is

$$\tau \cdot b_{ij} - (1-\tau)\cdot c_{ij} - \rho^e_{ij} t^e_{ij}.$$

Thus, no expropriation is weakly optimal whenever

$$\tau \leqslant \frac{c_{ij} + \rho^e_{ij} t^e_{ij}}{b_{ij} + c_{ij}}.$$

For a fixed penalty vector t^e, define

$$\tau^{\text{safe}}_{ij}(g; t^e) := \begin{cases} \min\left\{1, \dfrac{c_{ij} + \rho^e_{ij} t^e_{ij}}{b_{ij} + c_{ij}}\right\} & \text{if } i \in \mathrm{Arm}(g), \\ 1 & \text{if } i \notin \mathrm{Arm}(g). \end{cases}$$

Applying the same upper-bound argument as in Lemma 2, with $\tau^{\text{safe}}_{ij}(g; t^e)$ in place of $\tau^{\text{safe}}_{ij}(g)$, any obedient decision rule under penalties t^e satisfies

$$\mathbb{E}[TS(a, g; \theta)] \leqslant \sum_{k\in\mathcal{K}} V_k \min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\text{safe}}_{ij}(g; t^e).$$

Since $\tau^{\text{safe}}_{ij}(g; t^e)$ is weakly increasing in t^e_{ij}, this upper bound is maximized by setting $t^e_{ij} = \bar{t}_{ij}$ for every $i \neq j$. Hence every feasible rule satisfies

$$\mathbb{E}[TS(a, g; \theta)] \leqslant \sum_{k\in\mathcal{K}} V_k \min_{(i,j)\in\mathcal{Q}_k(g)} \tilde{\tau}^{\text{safe}}_{ij}(g).$$

This upper bound is attained by the nested no-expropriation decision rule from Proposition 1, with each baseline threshold $\tau^{\text{safe}}_{ij}(g)$ replaced by the modified threshold $\tilde{\tau}^{\text{safe}}_{ij}(g)$, and $t^e_{ij} = \bar{t}_{ij}$. Under this rule, posterior precision is capped at $\tilde{\tau}^{\text{safe}}_{ij}(g)$, so no expropriation deviation is profitable, and project k succeeds with probability $\min_{(i,j)\in\mathcal{Q}_k(g)} \tilde{\tau}^{\text{safe}}_{ij}(g)$. Therefore,

$$TS^{\text{partial}}(g) = \sum_{k\in\mathcal{K}} V_k \min_{(i,j)\in\mathcal{Q}_k(g)} \tilde{\tau}^{\text{safe}}_{ij}(g).$$

(iii) Contractible cooperation and shares: $TS^c(g) = TS(g)$. First, we show that $TS^c(g) \geqslant TS(g)$. Let σ^g be the nested no-expropriation decision rule constructed in the proof of Proposition 1 (see (A.2)–(A.3)). Under σ^g we have $\zeta_{ij} = 0$ for all $i \neq j$ almost surely, and each project k succeeds with probability $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$, so it achieves $TS(g)$. This decision rule is obedient and satisfies (A.22).

Second, we show that $TS^c(g) \leqslant TS(g)$. This inequality follows from the same steps as in the proof of Lemma 2. The binding restriction comes from the expropriation obedience constraints. Since expropriation actions remain non-contractible, the same upper bound continues to apply even when cooperative actions are contractible.

Finally, allowing the mediator to choose the shares $\{\alpha_{ik}\}$ does not affect the expected total surplus, which depends only on total project values. It also does not affect the obedience constraints (A.22): because cooperative actions are fixed in both terms, cooperative payoffs cancel when comparing deviations in expropriation. Hence, making shares contractible is immaterial for the welfare bound. □

A.4 Non-Contractible Communication: Cheap Talk

This appendix proves the results stated in Section 5.3. We first prove Proposition 3 regarding the cheap-talk game. We then show how to adapt the reasoning from the proof of Proposition 1 for the binary-state case to the continuum analogue.

A.4.1 Proof of Proposition 3

Proof. Fix an owner $g \in \mathcal{N}$. Because each target state is communicated to at most one armed party, every *risky* communication problem can be treated separately.

Consider first any task (i,j) with $i \notin \text{Arm}(g)$. Since party i is disarmed, sender j can truthfully reveal θ_j to i and task (i,j) succeeds with probability $1 = \tau_{ij}^{\text{safe}}(g)$.

Success and Failure Zones. Now consider any task (i,j) with $i \in \text{Arm}(g)$. To simplify notation, let $\tau^\star := \tau_{ij}^\star$ and $\delta := \delta_j$. Define the success zone and failure zone in $\mathbb{T} = [0,1)$ by $\mathbb{T}^{\text{success}} := [0,\tau^\star]$ and $\mathbb{T}^{\text{fail}} := (\tau^\star, 1)$.

Partition of the Success Zone. Let $s^\circ := \min\{2\delta, \frac{\tau^\star}{2}\}$. We partition $\mathbb{T}^{\text{success}}$ into $M := \lceil \tau^\star/(2\delta) \rceil$ contiguous intervals $S^{(1)}, \ldots, S^{(M)}$ such that

$$S^{(1)} := [0, s^\circ), \qquad S^{(M)} := [\tau^\star - s^\circ, \tau^\star], \qquad |S^{(m)}| \leqslant 2\delta \quad \text{for every } m,$$

and $\bigcup_{m=1}^{M} S^{(m)} = \mathbb{T}^{\text{success}}$. Write $S^{(m)} = [\nu^{(m-1)}, \nu^{(m)})$ with $\nu^{(0)} := 0$ for $m = 1, \ldots, M-1$ and $S^{(m)} = [\nu^{(m-1)}, \nu^{(m)}]$ with $\nu^{(M)} := \tau^\star$. Thus, $|S^{(m)}| = \nu^{(m)} - \nu^{(m-1)}$.

For each interval $S^{(m)}$, choose an intended receiver action $x^{(m)}$ such that

$$x^{(1)} = \delta, \qquad x^{(M)} = \tau^\star - \delta, \qquad x^{(m)} = \frac{\nu^{(m-1)} + \nu^{(m)}}{2} \quad \text{for } m = 2, \ldots, M-1$$

Partition of the Failure Zone. Define the linear mapping:

$$\phi(y) := \tau^\star + \omega \cdot y \quad \text{for } y \in \mathbb{T}^{\text{success}}, \quad \text{where } \omega := \frac{1-\tau^\star}{\tau^\star}.$$

Define failure intervals as follows:

$$F^{(1)} := (\tau^\star, \phi(\nu^{(1)})), \qquad F^{(M)} := \big[\phi(\nu^{(M-1)}), 1), \qquad F^{(m)} := \phi(S^{(m)}) \quad \text{for } m = 2, \ldots, M-1.$$

The sets $\{F^{(m)}\}_{m=1}^{M}$ are disjoint intervals that partition $\mathbb{T}^{\text{fail}}$, and

$$|F^{(m)}| = \omega|S^{(m)}| = \frac{1-\tau^\star}{\tau^\star}|S^{(m)}|.$$

Strategies. The sender's strategy is truthful with respect to this partition: send message m if and only if $\theta_j \in S^{(m)} \cup F^{(m)}$. Upon receiving message m, the receiver chooses the intended action $x_{ij} = \bar{x}_{ij} = x^{(m)}$ and no expropriation ($\zeta = \varnothing$).

Sender incentive compatibility. If $\theta_j \in S^{(m)}$, then by construction

$$\theta_j \in B(x^{(m)}, \delta) := \{\theta : d(x^{(m)}, \theta) \leqslant \delta\},$$

so the prescribed message yields task (i,j) success without expropriation.

If $\theta_j \in F^{(m)}$, then θ_j lies strictly outside every receiver success neighborhood. Under any message, the task fails:

$$B(x^{(m')}, \delta) \cap \big(\mathbb{T}^{\text{fail}}\big) = \varnothing \quad \text{for all } m'.$$

Thus, sender types in $F^{(m)}$ are indifferent, and the prescribed strategy is sequentially rational.

Receiver incentive compatibility. After receiving message m, the receiver's posterior is uniform on $S^{(m)} \cup F^{(m)}$. By construction of the failure zones, the relative measure of the success interval is

$$\frac{|S^{(m)}|}{|S^{(m)}| + |F^{(m)}|} = \frac{|S^{(m)}|}{|S^{(m)}| + \frac{1-\tau^\star}{\tau^\star}|S^{(m)}|} = \tau^\star.$$

Since the intended action $x^{(m)}$ covers all $S^{(m)}$ and none of $F^{(m)}$, the probability of success from choosing $x^{(m)}$ is $\tau^\star$.

We verify that no deviating action $y \in \mathbb{T}$ can yield a strictly larger probability of success. The success mass captured by any action y is $|B(y,\delta) \cap (S^{(m)} \cup F^{(m)})|$. If $B(y,\delta)$ intersects at most one of the two intervals, the captured mass is bounded by

$$|B(y,\delta) \cap (S^{(m)} \cup F^{(m)})| \leqslant \max\{|S^{(m)}|, |F^{(m)}|\} = |S^{(m)}|,$$

where the equality holds because $\tau^\star > 1/2$ implies $|F^{(m)}| < |S^{(m)}|$. Such a deviation yields a success probability of at most $\tau^\star$, making it unprofitable.

Suppose $M \geqslant 3$. This implies $\tau^\star > 4\delta$ and therefore $|S^{(1)}| = |S^{(M)}| = 2\delta$. For any inner message $m \in \{2, \ldots, M-1\}$, the distance between $S^{(m)}$ and $F^{(m)}$ in either direction around the circle is bounded below by 2δ. Thus, $B(y, \delta)$ cannot simultaneously intersect both $S^{(m)}$ and $F^{(m)}$, and for any inner cell $m \in \{2, \ldots, M-1\}$, any action captures at most mass $|S^{(m)}|$, meaning there is no profitable deviation from $x^{(m)}$.

For $M \geqslant 2$, consider $m \in \{1, M\}$ and a deviation $y \in \mathbb{T}$ such that $B(y, \delta)$ simultaneously intersects both $S^{(1)}$ and $F^{(1)}$. By symmetry, it suffices to analyze $m = 1$.

Intervals $S^{(1)}$ and $F^{(1)}$ are separated by two gaps on the circle $\mathbb{T}$. First, there is the counter-clockwise gap spanning the remainder of the success zone, with length $D_{cw} = \tau^\star - |S^{(1)}|$. If $D_{cw} \geqslant 2\delta$, then a radius-δ neighborhood cannot cover this gap and hence cannot simultaneously capture mass from both $S^{(1)}$ and $F^{(1)}$ across this gap. If $D_{cw} < 2\delta$, then such a neighborhood captures at most $2\delta - D_{cw} = |S^{(1)}| - (\tau^\star - 2\delta) < |S^{(1)}|$, where the last inequality uses $2\delta < \tau^\star$.

Second, there is the clockwise gap spanning the remainder of the failure zone, with length $D_{ccw} = 1 - \tau^\star - |F^{(1)}|$. An interval of length 2δ covering this gap captures at most $2\delta - D_{ccw} = 2\delta - 1 + \tau^\star + |F^{(1)}|$. For the deviation to be unprofitable, the captured mass must be bounded by the intended mass:

$$2\delta - 1 + \tau^\star + |F^{(1)}| \leqslant |S^{(1)}| \implies |S^{(1)}| - |F^{(1)}| \geqslant \tau^\star + 2\delta - 1.$$

Because $|F^{(1)}| = \omega |S^{(1)}|$, the following inequality must be satisfied:

$$|S^{(1)}| \geqslant \tau^\star \frac{\tau^\star + 2\delta - 1}{2\tau^\star - 1}. \tag{A.24}$$

By construction, $|S^{(1)}| = \min\{2\delta, \tau^\star/2\}$. If $|S^{(1)}| = \tau^\star/2$, then

$$\frac{\tau^\star}{2} \geqslant \tau^\star \frac{\tau^\star + 2\delta - 1}{2\tau^\star - 1} \iff \frac{1}{4} \geqslant \delta,$$

which is satisfied by Assumption $2^\star$ ($\delta \leqslant 1/4$). If $|S^{(1)}| = 2\delta$, then

$$2\delta \geqslant \tau^\star \frac{\tau^\star + 2\delta - 1}{2\tau^\star - 1} \iff (1 - \tau^\star)(\tau^\star - 2\delta) \geqslant 0,$$

which is satisfied by Assumption $2^\star$ ($\tau^\star > 1/2$ and $\delta \leqslant 1/4$).

Thus, the inequality (A.24) is satisfied, and no deviation can yield a higher expected success probability than $x^{(1)}$. By symmetric geometric construction, the same holds for $m = M$, concluding the proof of receiver incentive compatibility.

Probability of Success. In the constructed PBE, task (i, j) with $i \in \mathrm{Arm}(g)$ succeeds with probability $\tau^\star_{ij} = \tau^{\mathrm{safe}}_{ij}(g)$. □

A.4.2 The Mediator's Benchmark With Continuous State and Action Space

Stochastic Decision Rule. The proof of Proposition 1 extends to the continuum analogue after replacing the binary posterior precision with the *local posterior precision*:

$$q_{ij}(I_i) := \sup_{y \in \mathbb{T}} \mathbb{P}_\psi\big(d(y, \theta_j) \leqslant \delta_j \mid I_i\big), \qquad I_i = (\theta_i, a_i).$$

For an armed party i, the obedience constraint for the no-expropriation action implies $q_{ij}(I_i) \leqslant \tau^\star_{ij}$ at every information set. Thus, any productive task (i, j) can succeed with probability at most $\tau^{\mathrm{safe}}_{ij}(g)$. Hence, for any obedient decision rule with no expropriation,

$$\mathbb{P}_\psi(E_k) \leqslant \min_{(i,j) \in \mathcal{Q}_k(g)} \tau^{\mathrm{safe}}_{ij}(g),$$

and therefore,

$$\mathbb{E}_\psi[TS(a, g; \theta)] \leqslant \sum_{k \in \mathcal{K}} V_k \min_{(i,j) \in \mathcal{Q}_k(g)} \tau^{\mathrm{safe}}_{ij}(g).$$

The decision rule provided in Section 5.3 attains this upper bound. The obedience and optimality of refraining from expropriation follow the same logic as in Appendix A.1.

Deterministic Decision Rule. We show that in the special case where only one armed party receives information from others — which covers Examples 3 and 4 and all applications — there exists a deterministic decision rule attaining the upper bound on total surplus.

Proposition 7. *Fix an owner $g \in \mathcal{N}$. Suppose there exists a party $i \in \mathcal{N}$ such that every payoff-relevant task in $\bigcup_{k \in \mathcal{K}} \mathcal{Q}_k(g)$ is one of the following: (i) a self-task (ℓ, ℓ), (ii) a task (ℓ, j) performed by a disarmed party $\ell \notin \mathrm{Arm}(g)$, (iii) a task (i, j). There exists a deterministic obedient decision rule that attains the upper bound on total surplus, $TS(g)$.*

Proof. Let $\mathcal{J}_i(g) := \{j \in \mathcal{N} \backslash \{i\} : (i, j) \in \bigcup_{k \in \mathcal{K}} \mathcal{Q}_k(g)\}$ denote the set of targets party i must match across all projects. If $\mathcal{J}_i(g) = \varnothing$, the result is immediate. If $i \notin \mathrm{Arm}(g)$, the result is also immediate: all payoff-relevant non-self tasks are performed by disarmed parties and can be implemented by full revelation. Hence suppose $\mathcal{J}_i(g) \neq \varnothing$ and $i \in \mathrm{Arm}(g)$. Choose a bottleneck target $r \in \arg\min_{j \in \mathcal{J}_i(g)} \tau^{\mathrm{safe}}_{ij}(g)$. For each $j \in \mathcal{J}_i(g)$, let $\tau_j := \tau^{\mathrm{safe}}_{ij}(g)$ and $L_j := 2\delta_j/\tau_j$.

Bottleneck partition. Apply the construction in the proof of Proposition 3 to the task (i, r). This yields distinct intended actions $x_r^{(1)}, \ldots, x_r^{(M_r)} \in \mathbb{T}$, success-zone endpoints $0 = \nu^{(0)} <$

$\nu^{(1)} < \cdots < \nu^{(M_r)} = \tau_r$, and the associated success and failure intervals

$$S_r^{(m)} = \begin{cases} [\nu^{(m-1)}, \nu^{(m)}), & m = 1, \ldots, M_r - 1, \\ [\nu^{(M_r-1)}, \nu^{(M_r)}], & m = M_r, \end{cases}$$

$$F_r^{(1)} = (\tau_r, \phi(\nu^{(1)})), \quad F_r^{(M_r)} = [\phi(\nu^{(M_r-1)}), 1), \quad F_r^{(m)} = \phi(S_r^{(m)}) \quad \text{for } m = 2, \ldots, M_r - 1,$$

where $\phi(y) \coloneqq \tau_r + \omega_r y$ with $\omega_r \coloneqq (1 - \tau_r)/\tau_r$. Let $C_r^{(m)} \coloneqq S_r^{(m)} \cup F_r^{(m)}$. By construction,

$$\frac{|S_r^{(m)}|}{|C_r^{(m)}|} = \tau_r, \qquad S_r^{(m)} \subseteq B(x_r^{(m)}, \delta_r), \qquad \sup_{y \in \mathbb{T}} \frac{|B(y, \delta_r) \cap C_r^{(m)}|}{|C_r^{(m)}|} \leqslant \tau_r. \tag{A.25}$$

Coordination Device. Let $L_m^s \coloneqq \inf S_r^{(m)} = \nu^{(m-1)}$ and $L_m^f \coloneqq \inf F_r^{(m)}$. For each m, define $\eta_m : C_r^{(m)} \to [0, 1)$ by

$$\eta_m(y) \coloneqq \begin{cases} \dfrac{1}{2} - \dfrac{\tau_r}{2} + \dfrac{y - L_m^s}{|C_r^{(m)}|}, & y \in S_r^{(m)}, \\ \left(\dfrac{1}{2} + \dfrac{\tau_r}{2} + \dfrac{y - L_m^f}{|C_r^{(m)}|} \right) \pmod 1, & y \in F_r^{(m)}. \end{cases}$$

The map η_m is measure-preserving with respect to the normalized Lebesgue measure on $C_r^{(m)}$: on each piece it is affine with slope $1/|C_r^{(m)}|$, and $|S_r^{(m)}|/|C_r^{(m)}| = \tau_r$. Endpoint conventions for $S_r^{(m)}$ and $F_r^{(m)}$ affect η_m only on a Lebesgue-null set and do not affect the probability calculations that follow.

Deterministic rule. For every self-task (ℓ, ℓ), set $x_{\ell\ell} = \bar{x}_{\ell\ell} = \theta_\ell$. For every task (ℓ, j) with $\ell \notin \mathrm{Arm}(g)$, set $x_{\ell j} = \bar{x}_{\ell j} = \theta_j$. If $\theta_r \in C_r^{(m)}$, set $x_{ir} = \bar{x}_{ir} = x_r^{(m)}$, and for each $j \in \mathcal{J}_i(g) \backslash \{r\}$, set

$$x_{ij} = \bar{x}_{ij} \coloneqq \theta_j + L_j \left(\frac{1}{2} - \eta_m(\theta_r) \right) \pmod 1.$$

All expropriation actions are set to $\varnothing$. Remaining action components are set to arbitrary constants, and the decision rule $\bar{\sigma}^g(\cdot \mid \theta)$ places unit mass on this deterministic profile.

Party i's posterior. Conditional on $x_{ir} = x_r^{(m)}$, the state θ_r is uniform on $C_r^{(m)}$, so $\eta_m(\theta_r)$ is uniform on $[0, 1)$. For $j \in \mathcal{J}_i(g) \backslash \{r\}$, x_{ij} is a translation of θ_j by a function of $\eta_m(\theta_r)$. Since θ_j is independent of θ_r and uniformly distributed, x_{ij} is independent of $\eta_m(\theta_r)$. Thus, the vector of non-bottleneck recommendations does not reveal additional information about $\eta_m(\theta_r)$ or θ_r, and $\eta_m(\theta_r)$ remains uniform conditional on i's information $I_i = (\theta_i, a_i)$.

For each $j \in \mathcal{J}_i(g) \backslash \{r\}$, the posterior of $\theta_j \mid I_i$ is uniform on the arc of length L_j centered at

x_{ij}. Hence the local posterior precision is

$$q_{ij}(I_i) = \frac{2\delta_j}{L_j} = \tau_j = \tau_{ij}^{\text{safe}}(g),$$

attained at $y = x_{ij}$. For $j = r$, (A.25) implies $q_{ir}(I_i) = \tau_r$, attained at $y = x_r^{(m)}$.

No expropriation. For every relevant target $j \in \mathcal{J}_i(g)$, party i's posterior precision equals $\tau_{ij}^{\text{safe}}(g)$. For irrelevant targets, party i receives only constant recommendations and its posterior precision remains at the prior level, which is below the safety threshold by Assumption $2^\star$. Thus, refraining from expropriation is weakly optimal for all targets. Any other armed party receives only self-task or constant recommendations, so uninformed expropriation is unprofitable by the same assumption.

Project success. Task (i, r) succeeds if and only if $\theta_r \in S_r^{(m)}$, equivalently $|\eta_m(\theta_r) - \frac{1}{2}| \leqslant \tau_r/2$. For $j \in \mathcal{J}_i(g)\backslash\{r\}$, task (i, j) succeeds iff $d(x_{ij}, \theta_j) \leqslant \delta_j$, equivalently $|\eta_m(\theta_r) - \frac{1}{2}| \leqslant \tau_j/2$. Because $\tau_r \leqslant \tau_j$ for all $j \in \mathcal{J}_i(g)$, the success events are nested: whenever task (i, r) succeeds, every other task (i, j) with $j \in \mathcal{J}_i(g)$ also succeeds. Let $\mathcal{J}_{i,k}(g) := \{j \in \mathcal{J}_i(g) : (i, j) \in \mathcal{Q}_k(g)\}$. If $\mathcal{J}_{i,k}(g) = \varnothing$, every task in project k succeeds with probability one. Otherwise,

$$\mathbb{P}_{\bar{\psi}}(E_k) = \min_{j \in \mathcal{J}_{i,k}(g)} \tau_j = \min_{(\ell,j) \in \mathcal{Q}_k(g)} \tau_{\ell j}^{\text{safe}}(g),$$

where $\bar{\psi}(a, \theta) = p(\theta)\bar{\sigma}^g(a \mid \theta)$. Moreover, no alternative productive action can make task (i, j) succeed with probability greater than $q_{ij}(I_i) = \tau_j$. Therefore, for any deviation in productive actions, the probability that project k succeeds is bounded above by $\min_{j \in \mathcal{J}_{i,k}(g)} \tau_j$, which is the probability attained by the prescribed actions. Since project values are nonnegative, the prescribed productive actions are obedient.

Total surplus. Summing over projects,

$$\mathbb{E}_{\bar{\psi}}[TS(a, g; \theta)] = \sum_{k \in \mathcal{K}} V_k \min_{(\ell,j) \in \mathcal{Q}_k(g)} \tau_{\ell j}^{\text{safe}}(g),$$

which matches the upper bound established for the stochastic decision rule. □

A.5 Optimal Decision Rule Under Arbitrary Expropriation Costs

This appendix extends the characterization of the optimal decision rule to the case of arbitrary expropriation costs, under the assumption that only one armed party receives information about others.

Let $\mathcal{J}_i(g) := \{j \in \mathcal{N}\backslash\{i\} : (i, j) \in \bigcup_{k \in \mathcal{K}} \mathcal{Q}_k(g)\}$ denote the set of targets whose states the receiver must match, and let $\mathcal{J}_{i,k}(g) := \{j \in \mathcal{J}_i(g) : (i,j) \in \mathcal{Q}_k(g)\}$ denote the subset required for project k.

For each target $j \in \mathcal{J}_i(g)$, define the contribution to the total surplus of an expropriation attempt that succeeds with probability q:

$$e_{ij}(q) := q(b_{ij} - \lambda_{ij}) - (1-q)c_{ij}.$$

For each subset $R \subseteq \mathcal{J}_i(g)$, interpreted as the set of targets the receiver is recommended to expropriate, define

$$\begin{aligned} W_R(g) := \max_{\{q_j\},\{m_k\}} \quad & \sum_{k\in\mathcal{K}} V_k m_k + \sum_{j\in R} e_{ij}(q_j) \\ \text{subject to} \quad & q_j \in [\tau^\star_{ij}, 1] \quad \text{for all } j \in R, \\ & q_j \in [1/2, \tau^\star_{ij}] \quad \text{for all } j \in \mathcal{J}_i(g)\backslash R, \\ & 0 \leqslant m_k \leqslant 1 \quad \text{for all } k \in \mathcal{K}, \\ & m_k \leqslant q_j \quad \text{for all } k \in \mathcal{K} \text{ and } j \in \mathcal{J}_{i,k}(g). \end{aligned} \tag{A.26}$$

When $\mathcal{J}_{i,k}(g) = \varnothing$, the constraint $m_k \leqslant q_j$ is absent, and $m_k = 1$ at the optimum.

Proposition 8 (Arbitrary expropriation costs with a single armed receiver)**.** *Fix an owner $g \in \mathcal{N}$. Suppose there exists an armed party $i \in \mathcal{N}$ such that every payoff-relevant task in $\bigcup_{k\in\mathcal{K}} \mathcal{Q}_k(g)$ is one of the following: (i) a self-task (ℓ, ℓ), (ii) a task (ℓ, j) performed by a disarmed party $\ell \notin \mathrm{Arm}(g)$, (iii) a task (i, j). Then the maximum attainable total surplus is*

$$TS(g) = \max_{R\subseteq\mathcal{J}_i(g)} W_R(g),$$

and this value is attained by a nested decision rule that uses a single randomization device to correlate task successes across the receiver's targets.

Proof. We first establish the upper bound and then construct an obedient decision rule that attains it.

Upper bound. Fix any obedient decision rule σ and the associated joint distribution $\psi(a,\theta) := p(\theta)\sigma(a \mid \theta)$. Consider an on-path information set of the receiver i. For each target $j \in \mathcal{J}_i(g)$, let q_j denote the receiver's posterior precision about θ_j, and let $R \subseteq \mathcal{J}_i(g)$ denote the set of targets the receiver is recommended to expropriate at this information set.

Obedience in expropriation implies: if $j \in R$, then $q_j \geqslant \tau^\star_{ij}$; if $j \notin R$, then $q_j \leqslant \tau^\star_{ij}$. Posterior mode gives $q_j \geqslant 1/2$. Hence $q_j \in [\tau^\star_{ij}, 1]$ for $j \in R$ and $q_j \in [1/2, \tau^\star_{ij}]$ for $j \notin R$.

Project k succeeds only if the receiver succeeds in every task (i,j) with $j \in \mathcal{J}_{i,k}(g)$, and each such task succeeds with probability at most q_j. Therefore, the project-k success probability m_k satisfies $m_k \leqslant \min_{j \in \mathcal{J}_{i,k}(g)} q_j$, with $\min \varnothing := 1$. The contribution of expropriation attempts to total surplus is at most $\sum_{j \in R} e_{ij}(q_j)$.

The total surplus at this information set is thus bounded above by the value of the program (A.26) defining $W_R(g)$. Taking expectations over information sets and maximizing over R,

$$\mathbb{E}_\psi[TS(a, g; \theta)] \leqslant \max_{R \subseteq \mathcal{J}_i(g)} W_R(g).$$

Attainability. Let $R^* \in \arg\max_R W_R(g)$ and let (q^*, m^*) solve (A.26) at R^*. Because the objective is increasing in each m_k, the optimum sets m_k to its upper bound:

$$m_k^* = \min_{j \in \mathcal{J}_{i,k}(g)} q_j^*,$$

with $m_k^* = 1$ when $\mathcal{J}_{i,k}(g) = \varnothing$.

Let $\xi \sim \text{Unif}[0, 1]$ be a common randomization device. The mediator recommends the receiver's productive actions as

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j & \text{if } \xi \leqslant q_j^*, \\ -\theta_j & \text{if } \xi > q_j^*, \end{cases} \qquad j \in \mathcal{J}_i(g).$$

For $j \in R^*$, the mediator recommends $\zeta_{ij} = x_{ij}$; for $j \notin R^*$, it recommends $\zeta_{ij} = 0$. All self-tasks and tasks performed by disarmed parties are set equal to the true state and succeed with probability one. Payoff-irrelevant cooperative actions are set arbitrarily.

Posteriors and obedience. Conditional on its recommendation, the receiver's posterior precision about θ_j equals q_j^*. Thus for $j \in R^*$, $q_j^* \geqslant \tau_{ij}^\star$, and the recommended expropriation attempt is weakly optimal. For $j \notin R^*$, $q_j^* \leqslant \tau_{ij}^\star$, and no expropriation is weakly optimal.

The productive actions are also obedient. The recommended value for each target is the posterior mode, so following is weakly optimal task-by-task. Moreover, because all targets use the same ξ, task successes are nested: task (i, j) succeeds if and only if $\xi \leqslant q_j^*$. Hence project k succeeds with probability $\min_{j \in \mathcal{J}_{i,k}(g)} q_j^* = m_k^*$ when $\mathcal{J}_{i,k}(g) \neq \varnothing$, and with probability one otherwise. No deviation in productive actions can raise any task's success probability beyond q_j^*, so project-k success cannot exceed m_k^* under any deviation. Since project values are nonnegative, the prescribed productive actions are obedient.

Total surplus. The total surplus under this rule equals

$$\sum_{k \in \mathcal{K}} V_k m_k^* + \sum_{j \in R*} e_{ij}(q_j^*) = W_{R*}(g) = \max_{R \subseteq \mathcal{J}_i(g)} W_R(g),$$

matching the upper bound. □